



How it feels to be a dairy farmer: A cross-sectional survey of farm management and well-being of dairy farmers in Western Canada and Ontario

H. Tambadou,¹ B. Zwick,¹ A. Le Heiget,¹ B. Hagen,^{1,2} C. Winder,^{1,3} J. C. Plaizier,¹ J. Kinley,⁴ K. Ominski,¹ E. Pajor,^{1,5} H. Watson,⁶ and M. King^{1*}

¹Department of Animal Science, University of Manitoba, Winnipeg, MB R3T 2N2, Canada

²Canadian Centre for Agricultural Wellbeing, Cannington, ON L0E 1E0, Canada

³Department of Population Medicine, University of Guelph, Guelph, ON N1G 2W1, Canada

⁴Department of Clinical Health Psychology, University of Manitoba, Winnipeg, MB R3A 1R9, Canada

⁵Department of Production Animal Health, University of Calgary, Calgary, AB T2N 4N1, Canada

⁶Farm Management Canada, Ottawa, ON K2P 0W6, Canada

ABSTRACT

Dairy farming is widely recognized as a physically and psychologically demanding profession, yet limited quantitative evidence describes the well-being of Canadian dairy producers. This exploratory, hypothesis-generating, cross-sectional study investigated univariable associations between farm management factors—including housing systems, milking technologies, daily responsibilities, and social environments—and the physical, mental, and social well-being of 115 dairy farmers in Western Canada and Ontario. The survey collected detailed information on demographics, work responsibilities, work-life balance, social support, management practices, specific stressors, and physical health, alongside validated psychometric scales (Perceived Stress Scale, Generalized Anxiety Disorder-7, Patient Health Questionnaire-9, and Connor–Davidson Resilience Scale-10). Participants reported lower stress and anxiety levels and higher resilience compared with the broader Canadian farming population. However, substantial physical injuries, sleep deprivation, and stress related to financial and workload pressures were prevalent. Strong social networks, adequate rest, and manageable workloads were associated with more favorable mental-health outcomes, whereas financial and workload-related stressors were significantly associated with increased stress, anxiety, and depressive symptoms. These findings underscore the need for holistic, multisector strategies that integrate mental health support, labor and financial assistance, family-business planning, and safety initiatives to improve the sustainability and overall well-being of dairy farmers.

Key words: mental health, physical health, social support, work-life balance

INTRODUCTION

Farming is widely recognized as one of the most physically and psychologically challenging professions (Hounscome et al., 2012; Hassard and Teoh, 2013). Farmers face a multitude of obstacles that significantly affect their well-being, including long hours, heavy physical demands, financial stress, and unpredictable weather conditions (Fraser et al., 2005; Finnigan, 2019; Jones-Bitton, 2019). The stigma surrounding mental health in rural areas only exacerbates these issues (Polain et al., 2011; Wilton, 2019; Contzen and Häberli, 2021). This concern is reflected in the higher rates of mental health disorders and suicides among farmers compared with similar socioeconomic groups (Torske et al., 2016; Rudolphi et al., 2020; Jones-Bitton et al., 2020; Hagen et al., 2021).

Researchers have shown that certain agricultural sectors experience greater levels of stress, with dairy farming often regarded as one of the most taxing industries globally (Lunner Kolstrup et al., 2013), particularly in countries such as Ireland (Brennan et al., 2022) and the United Kingdom (Deary et al., 1997). Kato et al. (2022) highlighted concerns regarding the mental health of dairy farm managers in Japan, stemming from numerous challenges linked to running a dairy operation. Factors that may affect the mental well-being of dairy farmers encompass animal health and disease outbreaks, climatic conditions, regulatory policies, and economic volatility (Finnigan, 2019). In the dairy sector, ensuring high milk quality and the health of livestock are crucial priorities that necessitate careful management of cow nutrition and metabolic health. Maintaining proper hygiene for dairy cows is

Received October 24, 2025.

Accepted March 17, 2026.

*Corresponding author: meagan.king@umanitoba.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

vital for preventing infections and promoting udder health (Kurihara and Shimoura, 2008). Poor care can lead to milk contamination from pathogenic bacteria on the udders and teats. Dairy farmers must juggle these various challenges, balancing the financial viability of their business while ensuring their cows are housed appropriately and delivering high-quality milk (Kato et al., 2022). This balancing act demands significant physical and mental effort, often leading to personal, social, and financial sacrifices that can heighten stress and increase the risk of depression and anxiety.

Given that dairy production contributes around Can\$16 billion (approximately US\$12 billion) each year to the Canadian economy (Government of Canada, 2022), a deeper understanding of dairy farmers' well-being is crucial for the sustainability of the industry. Currently in Canada, comprehensive data on the health and well-being of farmers, as well as the primary stressors they face, is notably lacking (Jones-Bitton, 2019; Mendly-Zambo, 2023; Agriculture and Agri-Food Canada, 2025). Mixed-methods research has begun to shed light on the range of stressors experienced by Canadian farmers (Hagen et al., 2021; Thompson et al., 2022). National-level surveys have provided important baseline indicators of mental health in the agricultural population (Jones-Bitton et al., 2020; Thompson et al., 2022); however, these studies do not capture detailed, dairy-specific data about farm management or stressors, nor do they integrate related dimensions such as physical injury, work-life balance, or social support networks. King et al. (2021) examined several factors potentially associated with mental health in dairy farming in Ontario, Canada, including management practices, milk production, milk quality, and cow welfare, but their analysis was limited to farms using robotic milking systems. In contrast, Canadian dairy operations encompass a diverse range of housing configurations, barn designs, milking technologies, and management approaches, reflecting the heterogeneity of the sector. This absence of comprehensive, system-inclusive data limits the development of targeted interventions and hinders efforts to adequately address the unique challenges faced by dairy producers.

Our aim was to quantitatively assess the physical, mental, and social well-being of farmers in Western Canada and Ontario, while also exploring factors associated with their well-being. Key factors included housing conditions, management practices, farm responsibilities, work-life balance, social support networks, and financial and succession planning. Additionally, we looked more closely at the specific stressors encountered by dairy farmers in Canada. This study was designed to be exploratory and hypothesis-generating as opposed to testing any specific hypotheses.

MATERIALS AND METHODS

Study Design

A cross-sectional, observational study was carried out from May 2023 to April 2024, during which dairy farmers were invited to complete an online questionnaire using the Qualtrics survey platform (Qualtrics, 2005). Participants were recruited from British Columbia, Alberta, Saskatchewan, Manitoba, and Ontario; these provinces were selected because of language, logistics, and funding reasons. To qualify for inclusion, participants had to be at least 18 years old and consider themselves to be either a primary or secondary or co-decision-maker on a commercial dairy farm within the targeted provinces. As an honorarium for their time, participants were offered the option of receiving a 15 CAD Tim Hortons gift card.

The study protocol received approval from the Research Ethics Board at the University of Manitoba (HE2022-1459), and informed consent was obtained from all participants before starting the questionnaire and throughout the study. Participants were allowed to withdraw from the study at any time. Recruitment efforts involved provincial dairy boards, feed companies, and veterinarians disseminating information about the study to farmers. Farmers were also contacted via phone and email using contact information garnered from publicly available farm websites. Study advertisement also took place through farm events, tours, and *The Milk Producer* magazine, which is available online and in print. In addition, our study was promoted on social media platforms (Twitter [now X], LinkedIn, and Facebook) and shared by other organizations and individuals to enhance visibility and participation. Interested participants were invited to complete a recruitment survey to provide their farm information and email address, which allowed us to send them a link to the full survey. This approach ensured that we were engaging with legitimate participants before distributing the full survey link. After the study was complete, interested participants received an aggregated summary report of our findings.

Surveying Mental and Physical Health

Our survey (supplemental material: "Survey Questions for Dairy Farmers"; see Notes) employed validated psychometric questionnaires to evaluate stress, anxiety, depression, and resilience, and included questions related to occupational injuries and farm management practices. These questionnaires have previously been used to investigate the mental health of Canadian farmers in various commodity sectors, including dairy, beef, and pig farming (Jones-Bitton et al., 2020; King et al., 2021; Thompson et al., 2022).

To assess stress, the Perceived Stress Scale (PSS), a 10-item self-report questionnaire, was used to gauge individuals' perceptions of general life stress and their coping mechanisms (Cohen et al., 1983). The PSS is widely recognized for its effectiveness in measuring stress levels and identifying individuals at risk of certain psychiatric disorders (Cohen et al., 1983). Respondents rated their feelings or thoughts over the previous month using a 5-point Likert scale ranging from 0 (never) to 4 (very often). Higher scores indicate elevated stress levels, with scores categorized as follows: 0 to 13 representing low stress, 14 to 26 indicating moderate stress, and 27 to 40 signifying high stress (Cohen et al., 1983).

To assess the severity of anxiety symptoms, the Generalized Anxiety Disorder-7 (GAD-7) was employed. This scale is not a diagnostic tool but serves as an initial screening instrument for generalized anxiety disorder (Spitzer et al., 2006). Participants rated the frequency with which they experienced specific issues over the past 2 wk on a 4-point Likert scale, ranging from 0 (not at all) to 3 (nearly every day). The GAD-7 scores spanned from 0 to 21, with categorization as follows: none or minimal (0–4), mild (5–9), moderate (10–14), or severe (15–21). A score of 10 or higher is considered a reasonable cutoff for identifying generalized anxiety disorder (Spitzer et al., 2006).

To assess the severity of depressive symptoms, the Patient Health Questionnaire-9 (PHQ-9) was used. It is a recommended instrument for primary care settings to assess depression severity and monitor treatment efficacy (Kroenke et al., 2001). The PHQ-9 demonstrates strong internal consistency, contributing to its effectiveness in identifying potential cases of depression (Kroenke et al., 2001). Participants rate the frequency with which they have experienced specific depressive symptoms over the past 2 wk on a scale from 0 (not at all) to 3 (nearly every day). The scoring ranges from 0 to 27, categorizing the severity of depressive symptoms as none (0–4), mild (5–9), moderate (10–14), moderately severe (15–19), or severe (20–27). A score of 10 or higher is considered a reasonable threshold for identifying major depressive disorder (Kroenke et al., 2001; Spitzer et al., 2006).

The refined 10-question Connor–Davidson Resilience Scale (CD-RISC) was used to assess resilience, defined as one's ability to adapt to changes or problems as they arise. This scale focuses on participants' persistence and hardiness amidst adversity (Connor and Davidson, 2003; Campbell-Sills and Stein, 2007). Respondents provided ratings based on their agreement with 10 resilience-related statements from the past month, employing a 5-point Likert scale where 0 signifies "not true at all the time" and 4 indicates "true nearly all the time." The scores range from 0 to 40, with higher values representing greater resilience.

Additionally, physical and occupational health were evaluated by having participants report injuries sustained while working on their farms. They selected from options for types of injuries or health issues, such as "broken bones," "bruises/contusions," "respiratory diseases," and "chronic joint pain," among others (supplemental material: Survey Questions). Furthermore, participants were questioned about the causes of their injuries, with response options including "machinery or equipment," "work environment," and "animal-related," among others. Last, the quality and satisfaction of their sleep were also assessed (supplemental material: Survey Questions).

A group of researchers, farmers, and representatives from provincial milk boards participated in the review of the survey and offered valuable feedback. This collaboration ensured that the survey aligned with the needs of the dairy industry and met our research objectives effectively. Before deployment, the questionnaire underwent pretesting to evaluate clarity, relevance, and technical performance. The research team first conducted an internal expert review, followed by pilot testing with a small group of dairy farmers and agricultural professionals who completed the survey on their own devices and provided feedback on wording, flow, and functionality. Based on this input, the survey was refined by clarifying ambiguous items, adjusting response options, improving skip-logic, resolving formatting and technical issues in Qualtrics, and streamlining sections to reduce respondent burden. These revisions ensured that the final version was user-friendly, context-appropriate, and functioned reliably across platforms.

Surveying Farm Management and Life on the Farm

The survey took ~20 min to complete and included 6 primary sections: demographics, farming responsibilities, work-life balance, social environment and support, animal housing and management, and specific stressors (supplemental material: Survey Questions). Participants were allowed to complete the survey at their convenience and not necessarily in one sitting; they could close and re-open the survey as many times as necessary before submitting. Questions were organized into short, thematically coherent sections to maintain a logical flow and minimize cognitive burden, and skip-logic was used to ensure respondents only saw items relevant to their farm structure or role, reducing unnecessary question load. Pretesting with farmers and agricultural professionals also helped identify items that were overly long, repetitive, or unclear, leading to the removal or rewording of such questions to streamline the final instrument. In addition, progress indicators and clear section headers were included to help participants gauge their advancement

through the survey. Collectively, these measures helped maintain response quality and supported full completion, despite the survey's length. All participants who initiated the survey successfully completed it before the designated closure date. As a result, there were no instances of missing data related to the survey length.

Participants responded to questions about their farm management, including the number of years they had managed the farm, their regular tasks, and the proportion of their time spent working alone. Additionally, they provided insights regarding their ability to maintain a work-life balance, take vacations, find time for rest and recreation, and enjoy activities that bring them joy. Other questions addressed the size of their operations, housing types, and management practices. Farmers were also asked about their future operational plans, succession planning, and financial records. To assess how often they were stressed about various stressors, we employed a Likert scale ranging from 1 (never) to 5 (very often). These stressors encompassed financial concerns (e.g., farm finances, milk prices, quota value, and so on), weather-related issues (e.g., wildfires, flooding, drought), and industry-related challenges (e.g., worries about animal activists, public perception, emerging technologies).

Sample Size

There are ~4,500 dairy farms across Western Canada and Ontario (Government of Canada, 2023). We conducted post hoc sample size calculations with an allowable error of 5% and a confidence level of 95% using prevalence estimates from our study sample. A single cross-sectional survey can be used to measure prevalence because it offers a snapshot of how many people in a population are experiencing a certain condition during a specific period of time, which is consistent with standard epidemiological practice for estimating point prevalence (Kim, 2023; Xia et al., 2024). To assess stress, using a prevalence of 9% for high (≥ 27) perceived stress scores (Cohen et al., 1983), the needed sample size would be 121 participants. To assess anxiety or depression, using the prevalence of 2.6% for severe (≥ 15) anxiety scores (Spitzer et al., 2006) and 2.6% for moderately severe to severe (≥ 15) depression scores (Kroenke et al., 2001), the minimum required sample size would be 39 participants (Dean et al., 2013). Therefore, the sample of 115 farmers in this study is sufficient for estimating the prevalence of mental health challenges. However, it may not be robust enough for testing other, more exploratory analyses, such as differences in the prevalence of mental health outcomes based on housing systems, milking systems, and other factors.

Due to a large number of responses to the recruitment survey ($n = 3,687$), we used a Research Ethics Board-ap-

proved protocol to verify the legitimacy of responses. Responses were flagged based on the following criteria: (1) phone number area code was not Canadian and did not match the province or a neighboring province; (2) the IP address was from outside the 5 focal provinces; (3) there was no online presence to support farm's existence; (4) the survey responses were illogical or inconsistent with industry norms or contained incoherent computer language; (5) the Captcha score was less than 0.5; (6) contact with the respondent received no reply; and (7) the response came from an automated system (email address was a random string of characters), or more than 10 entries were received within 15 min from same email or IP address. Responses were considered ineligible if more than half of the criteria were met (4 of 7). If a response met fewer than 3 of the flag indicators, they were contacted to confirm the information provided in the recruitment survey. We initiated outreach to 13 participants for confirmation, and their responses were ultimately excluded from our analysis. To minimize the risk of duplicate survey submissions, several validation measures were implemented within Qualtrics. First, participants were required to enter the email address to which their survey link had been sent, allowing the system to automatically flag repeated attempts from the same individual. Additional built-in fraud-prevention features were enabled, including checks for extremely rapid completion times, unusually repetitive response patterns, and other suspicious metadata. Beyond platform-level controls, a secondary layer of custom validation was applied using embedded data fields. These included manual verification of participant information (e.g., email address, province of residence, and phone number when available), as well as cross-checking IP addresses and submission timestamps. Together, these procedures were designed to ensure data integrity by identifying and screening out potential duplicate or fraudulent responses before analysis.

To enhance the response rate of the survey, we ensured that each legitimate participant received up to 3 reminders before the survey was closed. We reached out to a total of 121 individuals, and 115 of them completed the survey successfully. The 6 potential participants who did not complete the survey were not included in the analysis. However, 2 of the 115 submissions were only 94% complete. The missing responses were about how concerned they felt regarding certain stressors, which were marked as "not applicable" in our dataset.

Statistical Analyses

Total scores for the outcome variables (PSS, GAD-7, PHQ-9, and CD-RISC-10) were calculated and categorized according to the detailed instructions provided for each scale, as reported in the methods of prior studies

(Cohen et al., 1983; Kroenke et al., 2001; Connor and Davidson, 2003; Spitzer et al., 2006; Campbell-Sills and Stein, 2007). Survey results were analyzed using means, SD, ranges, and proportions to assess physical health outcomes or prevalence of mental health outcomes, reflecting the percentage of a given population that exhibited specific characteristics over a particular period. Additionally, we explored mental and physical health outcomes in relation to key demographic factors, including age, province, and gender. All statistical analyses were conducted using RStudio software version 4.3.1 (Posit Team, 2024), with statistical significance set at $P \leq 0.05$ and tendencies declared at $P < 0.1$. Because the scores from the GAD-7 and PHQ-9 scales were not normally distributed, the data were square-root transformed, and normality was retested using the Shapiro–Wilk test and quantile-quantile plots. We used *t*-tests to compare means against previously published mental health scores. Analysis of variance was used to examine associations between mental health scores and survey variables. Chi-squared tests were employed to assess sleep variables. Only descriptive statistics were used to summarize the types of injuries and health problems that farmers sustained on their farms, as well as the causes of these injuries.

Effect sizes are reported alongside hypothesis tests to quantify the magnitude of observed associations. For *t*-tests, effect sizes were calculated using Cohen's *d* and interpreted using the conventional thresholds of 0.20 (small), 0.50 (medium), and 0.80 (large; Cohen, 1988). For ANOVA models, Cohen's *f* was used, with values of 0.10, 0.25, and 0.40 representing small, medium, and large effects, respectively (Cohen, 1988). For chi-squared analyses, Cramer's *V* was computed, and thresholds were interpreted following Cohen-derived guidelines, where for 2×2 tables 0.10, 0.30, and 0.50 denote weak, moderate, and strong associations, respectively (Cohen, 1988). These thresholds were derived from established effect-size conventions and adapted depending on the df.

RESULTS AND DISCUSSION

Study Population

In total, 115 farmers completed the survey. Table 1 shows the distribution by gender, age group, province, and decision-making role, as well as the number of years participants spent managing the farm. Most survey responses (83%) were collected between April and October 2023, with additional responses received between February and April 2024. This survey was conducted as part of a broader project, during which the research team visited a subset of farms to collect on-farm animal health data. To align recruitment efforts across both studies,

Table 1. Demographic representation of 115 dairy farmers from Western Canada and Ontario who completed surveys about their well-being and farm management practices

Variable	Number of farmers	Proportion (%)
Gender		
Women	35	30
Men	80	70
Age group (yr)		
18–24	3	3
25–34	31	27
35–44	43	37
45–54	17	15
55–64	18	16
≥65	3	3
Province		
Ontario	52	45
Manitoba	18	16
Saskatchewan	4	3
Alberta	26	23
British Columbia	15	13
Time spent managing the farm (yr)		
≥10	63	55
5–9	32	28
1–4	17	15
<1	2	2
Decision-making role		
Primary	60	52
Secondary/codecisionmaker	55	48

the survey invitation link was typically distributed one day after each farm visit. Because the Western Canadian farms were visited during the spring and summer months as part of the broader project, the majority of responses from these provinces were gathered between May and July 2023. In contrast, visits to Ontario farms occurred in September and October 2023, resulting in later response collection for that region. After a final recruitment effort in January 2024, additional responses were obtained between February and April 2024.

Comparisons for each scale to the population of Canadian farmers surveyed between February and May 2021 (Thompson et al., 2022) can be found in Table 2. Meanwhile, Tables 3 and 4 provide detailed information on the mean mental scores and physical outcomes for each demographic factor. The average score on the PSS among our farmers was 17.3 ± 6.8 (Table 2), which is significantly lower than a recent report mean of 18.7 ± 7.0 ($P = 0.04$; $n = 1,110$; Thompson et al., 2022). Among the surveyed participants, 24% reported low stress, 67% indicated moderate stress, and 9% were classified as experiencing high stress. The average resilience score was $28.1 (\pm 6.0)$, which is significantly higher than the Canadian farmer mean of 24.7 ± 6.2 ($P < 0.001$; $n = 980$; Thompson et al., 2022). Notably, ~70% of participants in our study scored below the general population average resilience score.

The average anxiety score of participants was 4.91 (± 4.26), significantly lower than Canadian farmer mean

of 6.12 ± 4.9 ($P = 0.01$; $n = 1,110$) for the GAD-7 as documented by Thompson et al. (2022). Approximately 16% of farmers scored higher than 10 on the GAD screening, compared with the 15% and 14% figures reported for Canadians by Mental Health Research Canada (MHRC) in spring and summer 2023, respectively. Moreover, 13% of participants experienced moderate anxiety symptoms, compared with the Canadian average of 10%. However, only 3% exhibited severe anxiety, while the national average was 5%. Further, a larger proportion of the farmers faced mild anxiety symptoms (32% compared with 24% in the general population), and those with minimal symptoms were fewer among the participants (52% vs. 62%).

The average depression score among our participants was $5.21 (\pm 4.35)$, which was not statistically different from the Canadian farmer mean of 5.89 ± 5.3 ($P = 0.2$; $n = 978$; Thompson et al., 2022). The prevalence of screening positive for depressive disorder, indicated by a PHQ-9 score greater than 10, was 16% in our study, compared with 23% of the overall Canadian population in 2023 (MHRC, 2023). When comparing our farmers to the Canadian average examined in the spring of 2023, a smaller percentage exhibited symptoms of minimal depressive disorder (49.6% vs. 54%), and a greater proportion (34.8% vs. 23%) was classified as experiencing mild symptoms. The prevalence of moderate symptoms was consistent at 13% across our participants and the general Canadian population. However, the prevalence of depression scores indicating moderately severe to severe symptoms in our participants was 2.6% compared with 12% for Canadians in general (MHRC, 2023).

Dairy farmers surveyed in the current study experienced significantly lower stress and anxiety scores compared with the average Canadian farmer, including across commodities (Thompson et al., 2022). Additionally, our participants scored notably higher on resilience assessments. However, no significant differences in depression scores existed between the 2 studies. It is important to note that our findings focus specifically on dairy farmers while Thompson et al. (2022) examined mental health of all farmers in the context of COVID-19. Therefore, we cannot conclusively determine whether these outcomes are a result of the post-COVID-19 landscape or if they are related to specific factors unique to the dairy sector. Further research is necessary to identify the underlying causes of these differences.

Notably, Canadians in general tended to register in the most severe categories, while farmers' scores fell mostly within the mild and moderate categories of anxiety (GAD-7) and depression (PHQ-9). Thompson et al. (2022) identified a similar pattern when they assessed the mental health scores of farmers compared to the general Canadian population during the COVID-19

Table 2. Comparison of the mental health scores of 115 dairy farmers in Western Canada and Ontario versus the average scores of Canadian farmers in a recent survey using *t*-tests

Variable	Range	Survey mean $\pm$ SD	Farmer mean $\pm$ SD ¹	<i>P</i> -value
PSS ²	0–33	17.3 ± 6.8	18.72 ± 7.03	0.04
GAD-7 ³	0–21	4.91 ± 4.26	6.12 ± 4.9	<0.001
PHQ-9 ⁴	0–24	5.21 ± 4.35	5.89 ± 5.3	0.2
CD-RISC 10 ⁵	10–40	28.1 ± 6.0	24.7 ± 6.2	<0.001

¹Thompson et al. (2022).

²Perceived Stress Scale (Cohen et al., 1983), which contains 10 questions relating to the respondents' feelings over the past month, scoring each question from 0 to 4 and summing them to get an overall score out of 40, where a greater score indicates a higher level of perceived stress.

³General Anxiety Disorder-7 (Spitzer et al., 2006), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 21, where a score of 10 or higher is considered a reasonable threshold for identifying cases of generalized anxiety disorder.

⁴Patient Health Questionnaire-9 (Kroenke et al., 2001), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 27, where a score of 10 or higher is considered a reasonable threshold for cases of depression.

⁵Connor–Davidson Resilience Scale-10 (Connor and Davidson, 2003), using 10 questions to assess resilience, or the ability of a person to cope or thrive despite hardships, over the past month. Each question had a score of 0 to 4, and scores were summed to get an overall score of 40, where a greater score indicates greater resilience.

pandemic. They found that more of the general population scored in the most severe categories of mental health issues. At the same time, a larger proportion of farmers fell into the moderate categories of anxiety and depression (Thompson et al., 2022). However, the prominent stigma toward mental health could lead to unintentionally underestimating the prevalence due to lower participation from individuals who are not ready to openly discuss or disclose information pertaining to their mental health (Torvik et al., 2012).

Regarding physical health, 82% of participants sustained work-related injuries or health issues. Farmers attribute these injuries to various factors, with machinery, working environment, and animals being the leading causes (22%). This is closely followed by injuries stemming from a combination of “working environment and animals” (17%) and injuries related solely to the “working environment” (14%). The data indicated that the most frequently reported injuries included major bruises (54%), chronic joint pain (45%, $n = 51$), lacerations (34%, $n = 39$), and bone fractures (32%, $n = 37$). The least reported health problems were amputations (2%), respiratory diseases (7%), and loss of hearing or vision (11%). The findings of this study raise substantial concerns regarding the physical health of dairy farmers, with 82% of participants reporting at least one work-related injury or health problem. Numerous studies document the demanding and hazardous nature of dairy farming

Table 3. Comparing mental health scores across demographic characteristics of 115 dairy farmers from Western Canada and Ontario using ANOVA and *t*-tests

Variable	Mental health scores (mean $\pm$ SD)				<i>P</i> -value			
	Stress (PSS) ¹	Anxiety GAD-7 ²	Depression PHQ-9 ³	Resilience CD-RISC ⁴	Stress	Anxiety	Depression	Resilience
Gender								
Women	18.5 (± 7.0) ^a	6.31 (± 5.48) ^a	6.40 (± 5.12) ^a	27.6 (± 6.7) ^a	0.2	0.04	0.08	0.5
Men	16.8 (± 6.6) ^a	4.30 (± 3.46) ^b	4.69 (± 3.89) ^b	28.3 (± 5.7) ^a	0.3	0.2	0.2	0.06
Age group (yr)								
18–24	14.7 (± 2.1) ^a	4.00 (± 5.29) ^a	3.67 (± 2.52) ^a	32.7 (± 1.5) ^a				
25–34	19.1 (± 6.8) ^a	5.84 (± 5.16) ^a	6.39 (± 5.51) ^a	27.5 (± 6.4) ^a				
35–44	17.8 (± 6.9) ^a	5.30 (± 3.46) ^a	5.42 (± 3.85) ^a	26.9 (± 5.9) ^a				
45–54	15.9 (± 6.5) ^a	5.00 (± 5.28) ^a	5.12 (± 4.68) ^a	27.4 (± 6.1) ^a				
55–64	15.3 (± 6.8) ^a	2.89 (± 2.74) ^a	3.33 (± 2.79) ^a	31.0 (± 4.9) ^a				
≥ 65	14.7 (± 2.1) ^a	2.33 (± 1.53) ^a	3.33 (± 1.53) ^a	33.3 (± 5.5) ^a				
Province					0.02	0.001	0.09	0.5
Ontario	18.0 (± 6.1) ^{ab}	5.12 (± 4.02) ^b	5.62 (± 3.91) ^{ab}	27.7 (± 5.9) ^a				
Manitoba	14.8 (± 7.4) ^b	3.61 (± 4.88) ^b	4.94 (± 5.70) ^{ab}	30.4 (± 5.5) ^a				
Saskatchewan	18.0 (± 6.1) ^{ab}	3.75 (± 3.50) ^b	5.00 (± 5.60) ^{ab}	27.2 (± 4.8) ^a				
Alberta	15.2 (± 6.6) ^b	3.35 (± 3.02) ^b	3.46 (± 3.93) ^b	27.8 (± 6.5) ^a				
British Columbia	21.4 (± 6.9) ^a	8.80 (± 4.18) ^a	7.20 (± 3.76) ^a	27.6 (± 6.6) ^a	0.6	0.04	0.2	0.5
Time spent managing the farm (yr)								
≥ 10	16.9 (± 6.3) ^a	4.30 (± 3.82) ^b	4.52 (± 3.51) ^a	28.1 (± 6.0) ^a				
5–9	18.0 (± 6.6) ^a	5.06 (± 3.92) ^{ab}	5.75 (± 4.57) ^a	27.6 (± 5.8) ^a				
1–4	18.2 (± 8.7) ^a	7.41 (± 5.61) ^a	7.06 (± 6.23) ^a	27.8 (± 6.0) ^a				
<1	10.5 (± 4.9) ^a	0	2.0 (± 2.8) ^a	35.5 (± 2.1) ^a	0.2	0.9	0.7	0.004
Decision-making role								
Primary	16.5 (± 7.1) ^a	4.90 (± 4.44) ^a	5.05 (± 4.60) ^a	29.6 (± 5.2) ^a				
Secondary/codecision-maker	18.2 (± 6.3) ^a	4.93 (± 4.08) ^a	5.38 (± 4.09) ^a	26.4 (± 6.4) ^b				

^{a,b}Mean mental health scores in the same column under the same table section with different superscripts are statistically different ($P < 0.05$) or tend to differ ($P < 0.01$).

¹Perceived Stress Scale (Cohen et al., 1983), which contains ten questions relating to the respondents' feelings over the past month, scoring each question from 0 to 4 and summing them to get an overall score out of 40, where a greater score indicates a higher level of perceived stress.

²General Anxiety Disorder-7 (Spitzer et al., 2006), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 21, where a score of 10 or higher is considered a reasonable threshold for identifying cases of generalized anxiety disorder.

³Patient Health Questionnaire-9 (Kroenke et al., 2001), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 27, where a score of 10 or higher is considered a reasonable threshold for cases of depression.

⁴Connor-Davidson Resilience Scale-10 (Connor and Davidson, 2003), using 10 questions to assess resilience, or the ability of a person to cope or thrive despite hardships, over the past month. Each question had a score of 0 to 4, and scores were summed to get an overall score of 40, where a greater score indicates greater resilience.

Table 4. Comparing sleep outcomes across demographic characteristics of 115 dairy farmers from Western Canada and Ontario using chi-squared analyses

Variable	Physical health outcomes					P-value	
	Getting enough sleep (%)		Sleeping hours (%)			Getting enough sleep	Sleeping hours
	No	Yes	7–8 h	5–6 h	<5 h		
Gender						0.06	0.5
Women	21	9	15	14	2		
Men	35	35	26	39	4		
Age (yr)						0.9	0.2
18–24	1	2	3	0	0		
25–34	16	11	10	16	0		
35–44	21	16	16	18	3		
45–54	8	7	7	7	1		
55–64	9	7	4	10	2		
≥65	2	1	0	0	1		
Province						0.1	0.6
Ontario	29	16	20	21	4		
Manitoba	6	9	6	10	0		
Saskatchewan	3	0	1	3	0		
Alberta	10	12	7	14	2		
British Columbia	7	6	7	6	0		
Time spent managing the farm (yr)						0.7	0.9
≥10	29	25	22	30	3		
5–9	17	10	11	15	2		
1–4	8	7	7	7	1		
<1	1	1	1	1	0		
Decision-making role						0.4	0.1
Primary	27	25	17	30	5		
Secondary/codecision-maker	29	19	23	23	1		

as one of the most dangerous sectors within agriculture. For example, Douphrate et al. (2013) reported elevated rates of work-related injuries and fatalities across dairy operations globally, emphasizing frequent exposure to multiple occupational hazards. Likewise, recent reviews described agriculture more broadly as a high-risk industry characterized by recurrent mechanical, environmental, and animal-related dangers (Douphrate et al., 2013; Qi et al., 2024). The primary sources of injury identified in our study—machinery, environmental conditions, and interactions with animals—are consistent with those cited in previous literature. Surveys of US farms and ranches, for example, attribute roughly 22% of injuries to livestock (Johnson et al., 2021), with some regions reporting even higher proportions. Other analyses similarly highlight animal-related incidents, machinery accidents, and slips or falls as major contributors to injury risk in dairy settings (Douphrate et al., 2013; Lindahl et al., 2013). The most commonly reported injury types—major bruises, chronic joint pain, lacerations, and fractures—also mirror patterns found in existing studies. Research on dairy working conditions frequently emphasizes the physical strain associated with repetitive tasks such as milking, which can lead to chronic musculoskeletal disorders, particularly in the upper limbs and back (Douphrate et al., 2014; Buisseret et al., 2024). Therefore, given the

persistent and multifaceted nature of occupational risks in dairy farming, there may be a need for further engineering controls, safety training, and more comprehensive occupational health monitoring.

According to Watson et al. (2015), adults must obtain at least 7 h of sleep each night to maintain optimal health. Over half of the farmers surveyed reported sleeping 5 to 6 h a night (53%, $n = 61$), while 6% reported sleeping less than 5 h. In comparison to the general Canadian adult population, a smaller proportion of farmers met the recommended sleep duration of 7 h per night (41% vs. 66%; Wang et al., 2022). Alarming, more than half of our participants reported failing to achieve this minimum sleep requirement. Additionally, more than half (56%) expressed dissatisfaction with their sleep quality. Sleep quality and duration play a critical role in maintaining mental health, emotional regulation, cognitive function, and overall psychological well-being (Vandekerckhove and Wang, 2017; Scott et al., 2021). Chronic sleep deprivation, defined as consistently obtaining fewer than 7 h per night, can have dire health implications, including weight gain, obesity, diabetes, hypertension, heart disease, stroke, depression, and increased mortality risk (Altevogt et al., 2006). For farmers who regularly operate heavy machinery and manage livestock (e.g., cattle), inadequate sleep presents an additional risk factor for

farm-related injuries (Shortall et al., 2018), which are notably higher in incidence, as evidenced by previous studies (Spengler et al., 2004; Heaton et al., 2010).

Demographics and Farmer Well-Being Outcomes

Among the participants, the age group with the highest representation was 35 to 44 yr, making up 37% of the total sample ($n = 43$). This was followed by the 25-to-34 age group at 27% ($n = 31$). Most farmers had been managing their operations for 10 yr or more (55%, $n = 63$), with those having 5 to 9 yr of experience closely following at 28% ($n = 32$). All 5 focal provinces were included, with Ontario representing the largest portion of participants at 49%. More detailed characteristics of the sample population are presented in Table 1.

The majority of participants (70%, $n = 80$) identified as men (see Table 1), which may have implications for our findings. There were no significant differences in stress or resilience scores between men and women (Table 3), and the associated effect sizes were small ($d = 0.25$ for stress and $d = 0.10$ for resilience). Women had significantly higher anxiety scores (6.3 ± 5.48) than men (4.3 ± 3.4 ; $P = 0.04$), with a medium effect size ($d = 0.55$). Women also tended to report higher depression scores (6.4 ± 5.1 vs. 4.69 ± 3.46 ; $P = 0.08$; Table 3), with a small effect size ($d = 0.37$). This finding aligns with established research in psychiatric epidemiology, which consistently indicates that women are more likely than men to experience anxiety disorders over their lifetimes (Regier et al., 1990; Bruce et al., 2005). A distinct difference emerged regarding injury causes and gender ($P = 0.01$), with men more frequently reporting injuries linked to machinery and the work environment. It is widely recognized that men may engage in riskier behaviors, resulting in higher injury rates and fatalities from such incidents (Udry, 1998). Many studies support the notion that men are more likely to endure farm-related injuries than women (McCurdy and Carroll, 2000; Hagel et al., 2004). These findings are consistent with the conclusions of Dimich-Ward et al. (2004), which highlighted that a larger number of men farmers sustained injuries involving farm machinery. Moreover, women tended to report a lower rate of getting enough sleep, with only 9% feeling well-rested compared with 35% of men. In addition to their off-farm employment, women actively participate in various farm tasks, such as caring for livestock, managing crop production, and maintaining equipment (Jansen et al., 2019; Elliott et al., 2022). Furthermore, they take on significant responsibilities at home, including childcare, meal preparation, and other domestic duties (Wheeler et al., 1983).

When looking at different age groups, farmers aged 65 and older showed a tendency toward higher resilience

scores ($P = 0.06$). However, no significant differences were found in perceived stress, anxiety, or depression scores across age groups. Notably, the effect sizes were medium across comparisons (Supplemental Table S6, see Notes), indicating differences of practical magnitude that may have gone undetected due to limited statistical power (e.g., small group sizes or high within-group variability). Therefore, older farmers may exhibit greater resilience, but age-related differences in stress, anxiety, and depression were not detected in our sample. Regarding physical health, all participants aged 65 and older reported having sustained work-related injuries. A significant number of individuals in the 35-to-44 age group (88%) and the 25-to-34 age group (77%) also reported work injuries. Previous research has shown that individuals with conditions like arthritis, rheumatism, or sensory impairments—such as hearing loss or reduced vision—are at an elevated risk for agricultural-related injuries (Lewis et al., 1998; McCurdy and Carroll, 2000). Age-related declines in reflex speed could also make older farmers more vulnerable to injuries (Etherton et al., 1991). However, it is important to note that the increased risk of injuries among older farmers may also be connected to their longer experience in the dairy industry because they may have faced a higher frequency of injuries throughout their careers.

A significant association was observed between anxiety scores and years of farm management experience ($P = 0.04$; $f = 0.31$). Farmers with ≥ 10 yr of management experience reported lower anxiety levels (4.3 ± 3.8) compared with those with 1 to 4 yr of experience, who exhibited higher anxiety scores (7.4 ± 5.6 ; $P = 0.05$). These findings suggest that greater management experience may confer a protective effect against anxiety severity. Interestingly, stress, depression, resilience scores, and sleep variables showed no significant differences based on experience. Additionally, the effect sizes for these comparisons were small (Cohen's $f = 0.16$ for stress, 0.24 for depression, and 0.18 for resilience) and the associations for sleep variables were considered weak as well (Cramer's $V < 0.2$).

Although resilience scores were consistent across provinces ($P = 0.5$; $f = 0.17$), there were significant associations with medium and large effect sizes for stress ($P = 0.02$; $f = 0.33$), anxiety ($P = 0.001$; $f = 0.43$), and depression scores ($P = 0.02$; $f = 0.27$). Farmers in British Columbia reported the highest average stress and anxiety scores, often coinciding with elevated depression scores. In contrast, farmers in Alberta and Manitoba recorded the lowest stress scores. These findings raise important concerns about the mental health challenges faced by farmers in British Columbia, likely influenced by several factors, including the high political, economic, and physical costs associated with dairy farming in the province, making it increasingly difficult for farmers to

sustain their operations. The interpretation of provincial differences should be made cautiously because survey responses were collected at different times of the year, with Western Canadian farmers responding mainly in spring and summer and Ontario farmers in the fall, periods that differ in workload and stress levels. These seasonal differences in response timing align with province-specific farming cycles and may have influenced outcomes because participants in western provinces responded during periods of intense spring and summer workloads, whereas Ontario farmers completed the survey during fall harvest and winter preparation periods. These seasonal demands could help explain why provinces such as British Columbia showed the highest stress, anxiety, and depression scores because these farmers responded during a time of high physical and economic pressure, which may influence provincial differences in well-being. In contrast, Ontario farmers' mental health measures may have captured the cumulative fatigue or financial uncertainty typical of late-season transitions rather than the acute labor intensity of early summer. Together, these timing differences suggest that mental health scores across provinces reflect not only structural or socioeconomic factors but also the seasonal rhythms of dairy production that shape stress, workload, and emotional strain at different points in the farming year. In addition, several provinces had small sample sizes, reducing statistical reliability and limiting the generalizability of provincial comparisons. Therefore, these provincial comparisons, as with all our univariate analyses, should be considered as exploratory and hypothesis-generating.

Regarding the role of farmers as either primary or secondary or co-decision-makers, there were no notable differences in stress, anxiety, depression scores, or physical health. However, primary decision-makers had greater resilience scores than secondary or co-decision-makers ($P = 0.004$) with a large effect size ($d = 0.5$). This increased resilience may stem from having more control over decision-making in their operations. In contrast, secondary decision-makers might feel more stressed because of a perceived lack of control, especially when the primary decision-makers' choices clash with their own values, leading to feelings of frustration.

Farm Responsibilities and Work-Life Balance

Our survey revealed that a significant majority of farmers, specifically 77% ($n = 88$), work 7 d a week, while a smaller group, representing 23%, work 5 or 6 d a week. All farmers we surveyed reported managing multiple daily tasks, the most common being animal care, barn cleaning, financial management, cow milking, and seasonal cropping. Approximately 47% of participants ($n = 54$) dedicated more than 60% of their time to activi-

ties related to animal care. Similarly, 47% of participants spent greater than 60% of their time working alone. Alarming, only 5% of farmers reported being able to take time to rest or engage in leisure activities every day, and half reported doing so only a few times a week. Additionally, 10% of farmers did not take any vacation throughout the year, and 28% took fewer than 6 d off annually. Just 31% felt they had achieved a healthy work-life balance, and the majority (53%) indicated they only occasionally reached this balance. We chose not to define work-life balance explicitly, acknowledging that farmers lead unique lifestyles with varying responsibilities. This approach allowed participants to interpret work-life balance based on their personal experiences, shedding light on the diverse realities that shape farmers' lives.

Further analysis revealed no significant associations between mental health scores and sleep outcomes based on factors such as time spent with animals, working alone, daily tasks, weekly workdays, and vacation time ($P > 0.1$; Supplemental Tables S1 and S2, see Notes). Despite the absence of significant differences, several of these factors demonstrated moderate effect sizes in relation to certain mental health indicators ($0.10 < f < 0.4$; Supplemental Table S6). For example, the association between daily tasks and stress scores yielded an effect size of $f = 0.32$, which is classified as a moderate effect. The presence of moderate effect sizes indicates that these relationships may still hold practical or clinical relevance. This pattern suggests that the study may have lacked sufficient statistical power to detect differences, or that the effects, although meaningful, were subtle and context-dependent. In particular, the moderate effect observed between daily tasks and stress ($f = 0.32$) implies that the nature or intensity of daily farm responsibilities may influence stress levels, even if this trend did not reach significance. These findings highlight the potential importance of occupational factors in shaping farmers' mental health and warrant further investigation using larger samples, more refined measures, and multivariable analyses.

Significant differences with large effect sizes ($f = 0.41$ for stress and $f = 0.31$ for resilience) were observed in perceived stress and resilience scores related to achieving a healthy work-life balance ($P < 0.001$; Table 5). Farmers who were able to maintain a healthy work-life balance reported lower perceived stress scores (13.7 ± 6.4) compared with those who felt they only occasionally achieved this balance (18.1 ± 6.0 , $P = 0.007$) or not at all (21.0 ± 7.1 , $P < 0.001$). Furthermore, individuals with a healthy work-life balance also showed higher resilience scores (30.4 ± 7.0) than those struggling to strike this balance (25.4 ± 5.8 ; $P = 0.01$). In our analysis of anxiety, stress, and depression scores, we found no significant differences, and only small effect sizes,

Table 5. Comparing work-life balance variables with mental health scores of 115 dairy farmers in Western Canada and Ontario using ANOVA and *t*-tests

Variable	n	Mental health outcomes (mean $\pm$ SD)				P-value			
		Stress (PSS) ¹	Anxiety (GAD-7) ²	Depression (PHQ-9) ³	Resilience (CD-RISC) ⁴	Stress	Anxiety	Depression	Resilience
Healthy work-life balance						<0.001	0.1	0.1	<0.001
Yes	35	13.8 (± 6.4) ^c	3.6 (± 4.6) ^a	4.3 (± 5.0) ^a	30.5 (± 7.0) ^a				
Sometimes	60	18.2 (± 6.0) ^{ab}	5.3 (± 4.1) ^a	5.0 (± 3.8) ^a	27.5 (± 5.0) ^{ab}				
No	19	21.0 (± 7.1) ^a	6.1 (± 3.7) ^a	7.4 (± 4.3) ^a	25.4 (± 5.8) ^b				
Time for rest, recreation, or something bringing joy						0.8	0.8	0.9	0.01
Every day	6	19.8 (± 7.4) ^a	5.17 (± 5.60) ^a	5.67 (± 2.16) ^a	26.2 (± 7.3) ^{abc}				
Nearly every day	25	14.9 (± 6.9) ^a	4.48 (± 4.30) ^a	5.08 (± 4.74) ^a	31.6 (± 6.6) ^a				
A few times during the week	58	17.0 (± 6.4) ^a	4.74 (± 3.64) ^a	5.09 (± 3.56) ^a	27.5 (± 5.2) ^{bc}				
Once/wk	13	18.7 (± 5.7) ^a	5.69 (± 6.22) ^a	6.08 (± 6.78) ^a	28.1 (± 5.2) ^{abc}				
<Once/wk	13	20.8 (± 7.6) ^a	5.62 (± 4.27) ^a	4.92 (± 5.06) ^a	25.2 (± 6.7) ^{bc}				
Number of days worked/wk						0.2	0.5	0.07	0.9
Every day	88	16.8 (± 7.0) ^a	4.65 (± 3.81) ^a	4.75 (± 3.97) ^a	28.2 (± 6.0) ^a				
≤ 6 d	27	18.9 (± 5.3) ^a	5.73 (± 5.57) ^a	6.54 (± 5.24) ^b	27.9 (± 6.3) ^a				
Vacation time						0.2	0.2	0.3	0.4
None	12	15.6 (± 9.1) ^a	3.67 (± 9.1) ^a	3.58 (± 3.29) ^a	29.3 (± 5.5) ^a				
1–2 d	16	16.1 (± 7.1) ^a	3.81 (± 2.95) ^a	4.19 (± 4.32) ^a	28.3 (± 6.8) ^a				
3–6 d	16	19.8 (± 8.5) ^a	6.25 (± 4.78) ^a	6.56 (± 4.40) ^a	26.1 (± 5.9) ^a				
1 wk	23	16.3 (± 7.7) ^a	5.61 (± 4.06) ^a	5.26 (± 4.75) ^a	29.1 (± 6.9) ^a				
2 wk	28	19.3 (± 4.7) ^a	5.64 (± 5.52) ^a	6.14 (± 5.09) ^a	26.9 (± 5.1) ^a				
>3 wk	20	15.7 (± 5.3) ^a	3.65 (± 2.91) ^a	4.55 (± 5.98) ^a	29.4 (± 6.0) ^a				

^{a–c}Mean mental health values in the same column under the same table section with different superscripts are statistically different ($P < 0.05$) or tend to be significant ($P < 0.1$).

¹Perceived Stress Scale (Cohen et al., 1983), which contains ten questions relating to the respondents' feelings over the past month, scoring each question from 0 to 4 and summing them to get an overall score out of 40, where a greater score indicates a higher level of perceived stress.

²General Anxiety Disorder-7 (Spitzer et al., 2006), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 21, where a score of 10 or higher is considered a reasonable threshold for identifying cases of generalized anxiety disorder.

³Patient Health Questionnaire-9 (Kroenke et al., 2001), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 27, where a score of 10 or higher is considered a reasonable threshold for cases of depression.

⁴Connor–Davidson Resilience Scale-10 (Connor and Davidson, 2003), using 10 questions to assess resilience, or the ability of a person to cope or thrive despite hardships, over the past month. Each question had a score of 0 to 4, and scores were summed to get an overall score of 40, where a greater score indicates greater resilience.

based on how often farmers set aside time for rest or recreational activities throughout the week, except in the case of resilience scores ($P = 0.01$; $f = 0.35$). Farmers who allocated time for rest or recreation nearly every day displayed higher resilience scores (31.6 ± 6.6) than those who engaged in these activities only a few times a week (27.4 ± 5.2 ; $P = 0.03$) or less than once a week (25.1 ± 6.7 ; $P = 0.01$). Additionally, having more time for rest and recreation was moderately associated ($V = 0.27$) with better sleep quality ($P = 0.04$). More farmers who successfully achieved a healthy work-life balance were getting sufficient sleep ($P < 0.001$; Table 6), and the association was moderate ($V = 0.4$). Although this research does not establish causation, it does demonstrate an association between balancing work and personal life and obtaining sufficient sleep. Having adequate time for oneself, family, and personal activities appears to foster a well-structured day, allowing for restful sleep. These findings suggest that long working hours may be normalized within the dairy

industry, which has a significant effect on the physical and mental well-being of farmers.

Most participants in our study worked daily, aligning with the Statistics Canada (2023) report, which shows that 80% of farm operators work more than 40 h a week. Such long working hours are often associated with health issues, including sleep deprivation, which can negatively affect overall well-being (Hale, 2005; Virtanen et al., 2009). Notably, Hale (2005) found that individuals working more than 50 h a week face a higher risk of sleeping less than 6.5 h per day compared with those with a work week of less than 35 h. Similarly, Kivimäki et al. (2015) identified a strong link between extended working hours and an elevated risk of heart attacks and strokes, particularly in lower socioeconomic groups. The mental health repercussions are equally alarming because consistent long hours correlate with higher rates of depression and anxiety (Kivimäki et al., 2015). Although our study did not provide clear evidence of significant links between the number of working days and mental health scores,

Table 6. Comparing work-life balance variables with sleep outcomes of 115 dairy farmers in Western Canada and Ontario using chi-squared analyses

Variable	Physical health outcomes (%)					P-value	
	Getting enough sleep		Sleeping hours			Getting enough sleep	Sleeping hours
	No	Yes	7–8 h	5–6 h	<5 h		
Healthy work-life balance						<0.001	0.2
Yes	10	21	15	15	1		
Sometimes	32	20	23	25	3		
No	14	3	3	12	2		
Time for rest, recreation, or something bringing joy						0.1	0.5
Every day	1	4	3	2	0		
Nearly every day	10	12	10	11	0		
A few times during the week	29	22	23	24	3		
Once/wk	8	3	3	8	0		
<Once/wk	9	3	1	8	3		
Number of days worked/wk						0.2	0.6
Every day	40	36	32	14	5		
≤6 d	16	7	8	39	1		
Vacation time						0.6	0.2
None	5	5	1	8	2		
1–2 d	6	8	7	6	1		
3–6 d	10	4	4	9	1		
1 wk	13	7	7	13	0		
2 wk	14	10	14	9	2		
>3 wk	8	10	8	9	1		

77% of farmers worked every day of the week, with 69% taking one week of vacation or less annually. Those who were able to balance their professional and personal lives and took more frequent breaks during the week experienced lower stress scores and greater resilience. Even though many farmers perceive relentless work as a sign of commitment, which is expected in the farming community, the personal and social tolls that come with it should not be ignored (Wilkie, 2010). The demanding nature of farming, alongside the pressure to be consistently available, can lead to heightened stress, social isolation, and burnout (Maslach, 2003; Chengane et al., 2021). This is especially relevant when considering the expectations of farmers to meet production requirements, milk quality standards, animal care regulations, and sustainability, among other factors. Such responsibilities can elevate psychological stress (Kato et al., 2022), significantly threatening their safety and overall well-being. Consequently, this stress can result in increased risks of injuries, feelings of loneliness, and decreased motivation (Wheeler et al., 1983; Rook, 1987). Many farmers find it challenging to take a break from their responsibilities or to secure trustworthy help, which makes it difficult for them to establish a healthy work-life balance (Walker and Walker, 1988).

Social Environment and Support

The average reported number of workers per farm was 6.64 (± 12.1). Most farmers, specifically 62% ($n = 71$), in-

dicated they worked alongside both family members and employees. In contrast, 34% ($n = 39$) worked exclusively with family, and a smaller portion, just 4% ($n = 4$), had nonfamilial employees in their workforce. Significant differences in anxiety ($P = 0.006$; Table 7) and depression ($P = 0.01$) were observed based on who worked on the farm, with medium effect sizes ($f = 0.34$ and $f = 0.32$, respectively). Notably, farmers who worked exclusively with family members showed significantly lower anxiety scores (3.94 ± 2.59 compared with 9.75 ± 7.5 ; $P = 0.04$) and depression scores (4.28 ± 3.42 versus 11.5 ± 9.3 ; $P = 0.007$) than their counterparts who worked solely with nonfamilial employees. Additionally, farmers who worked with both family and employees exhibited lower depression scores when compared with those who worked exclusively with nonfamilial employees ($P = 0.02$).

Fortunately, 83.3% ($n = 95$) of participants expressed having someone at work to confide in. When assessing satisfaction within their social networks, a larger percentage of farmers (30.4%, $n = 35$) indicated they were “very satisfied” with support at home, contrasting with only 19.1% ($n = 22$) who felt the same about work support. Conversely, an equivalent percentage (6.1%, $n = 7$) reported being “very dissatisfied” with home and work support. Farmers reporting having someone to talk to at work for support demonstrated lower anxiety (4.40 ± 4.00 vs. 7.00 ± 4.52 ; $P = 0.03$) than those lacking such support. The perceived quality of support at home emerged as a significant determinant of mental health. Farmers who reported being satisfied with their home support

Table 7. Comparing social environment variables with mental health scores of 115 dairy farmers in Western Canada and Ontario using ANOVA and *t*-tests

Variable	n	Mental health outcomes (mean $\pm$ SD)					P-value			
		Stress (PSS) ¹	Anxiety (GAD-7) ²	Depression (PHQ-9) ³	Resilience (CD-RISC) ⁴	Stress	Anxiety	Depression	Resilience	
Total workers										
Who works on the farm	4	20.8 (± 5.2) ^a	9.75 (± 7.5) ^a	11.5 (± 9.3) ^a	28.3 (± 8.3) ^a	0.1	0.2	0.3	0.07	
Employees	39	16.1 (± 7.0) ^a	3.95 (± 2.59) ^b	4.28 (± 3.42) ^{bc}	28.1 (± 5.9) ^a	0.3	0.006	0.01	0.8	
Family members and employees	71	17.7 (± 6.7)	5.04 (± 4.51) ^{ab}	5.34 (± 4.24) ^b	28.2 (± 6.1) ^a					
Someone to talk to at work										
Yes	95	16.7 (± 6.8) ^a	4.4 (± 4.0) ^b	4.79 (± 4.44) ^b	28.4 (± 5.5) ^a	0.1	0.004	0.07	0.4	
No	20	19.8 (± 6.2) ^a	7.0 (± 4.5) ^a	7.21 (± 4.44) ^a	26.7 (± 5.5) ^a					
Satisfied with support at home						<0.001	<0.001	0.003	0.02	
Very satisfied	22	14.9 (± 6.6) ^b	3.60 (± 3.21) ^{ab}	4.03 (± 4.69) ^a	30.4 (± 5.2) ^a					
Satisfied	48	15.3 (± 6.0) ^b	3.55 (± 2.66) ^{ab}	4.17 (± 3.41) ^{ab}	28.4 (± 5.5) ^{ab}					
Neutral	33	21.4 (± 5.6) ^{ab}	6.92 (± 5.18) ^a	7.00 (± 4.16) ^{ab}	26.3 (± 6.4) ^{ab}					
Dissatisfied	7	22.7 (± 7.3) ^a	8.57 (± 3.69) ^{ab}	7.57 (± 3.69) ^{ab}	26.1 (± 7.0) ^{ab}					
Very dissatisfied	2	22.2 (± 7.3) ^{ab}	8.60 (± 3.71) ^a	9.60 (± 5.55) ^b	22.8 (± 8.3) ^b					
Not applicable	2	22.0 ^{ab}	14.0 ^a	7.00 ^{ab}	23.0 ^{ab}					
Satisfied with support at work						0.08	0.5	0.9	0.01	
Very satisfied	22	13.9 (± 7.2) ^a	3.7 (± 4.8) ^a	4.00 (± 4.3) ^a	31.5 (± 5.3) ^a					
Satisfied	48	16.3 (± 6.3) ^a	4.40 (± 3.8) ^a	5.00 (± 3.86) ^a	28.3 (± 5.8) ^{ab}					
Neutral	33	20.1 (± 6.0) ^a	5.50 (± 3.87) ^a	5.50 (± 3.87) ^a	26.6 (± 5.8) ^b					
Dissatisfied	7	22.3 (± 4.7) ^a	9.1 (± 5.1) ^a	9.90 (± 4.30) ^a	22.7 (± 6.8) ^{ab}					
Very dissatisfied	2	18.5 (± 10.6) ^a	5.5 (± 3.5) ^a	7.00 (± 7.07) ^a	29.0 (± 5.7) ^b					
Not applicable	3	15.3 (± 6.5) ^a	5.0 (± 7.8) ^a	3.00 (± 3.61) ^a	28.3 (± 4.7) ^{ab}					
Satisfied with veterinarian						0.6	0.2	0.2	0.3	
Very satisfied	60	17.1 (± 7.5) ^a	4.93 (± 4.05) ^a	5.13 (± 4.0) ^a	28.3 (± 6.2) ^a					
Satisfied	35	16.4 (± 6.1) ^a	3.89 (± 3.04) ^a	4.23 (± 3.96) ^a	28.3 (± 4.9) ^a					
Neutral	3	18.3 (± 4.2) ^a	5.00 (± 2.0) ^a	6.33 (± 1.15) ^a	21.0 (± 2.6) ^a					
Dissatisfied	1	17.0	7.00	10.0	34.0					
Very dissatisfied	0	—	—	—	—					
Not applicable	16	19.8 (± 5.1) ^a	6.94 (± 6.69) ^a	7.13 (± 6.11) ^a	27.9 (± 7.6) ^a					
Satisfied with feed representative						0.02	0.07	0.04	0.02	
Very satisfied	57	15.9 (± 7.2) ^{ab}	4.49 (± 4.37) ^{ab}	4.44 (± 4.55) ^a	29.7 (± 4.9) ^a					
Satisfied	40	17.9 (± 5.7) ^{ab}	4.55 (± 2.99) ^{ab}	5.38 (± 3.55) ^a	26.6 (± 4.9) ^a					
Neutral	2	23.5 (± 13.4) ^a	8.50 (± 0.71) ^a	10.5 (± 4.9) ^a	25.5 (± 4.9) ^a					
Dissatisfied	0	—	—	—	—					
Very dissatisfied	0	—	—	—	—					
Not applicable	15	20.9 (± 4.8) ^a	7.33 (± 5.9) ^a	7.33 (± 4.61) ^a	25.5 (± 7.6) ^a					
Satisfied with equipment supplier						0.3	0.6	0.02	0.3	
Very satisfied	23	15.4 (± 7.7) ^a	4.0 (± 5.0) ^a	4.1 (± 2.9) ^{ab}	29.2 (± 5.3) ^{ab}					
Satisfied	34	16.3 (± 7.4) ^a	4.2 (± 4.2) ^a	3.7 (± 4.1) ^{ab}	28.4 (± 6.1) ^{ab}					
Neutral	8	16.8 (± 5.7) ^a	5.4 (± 5.8) ^a	5.3 (± 4.4) ^{ab}	27.4 (± 6.1) ^{ab}					
Dissatisfied	5	17.8 (± 3.8) ^a	7.0 (± 4.9) ^a	8.2 (± 4.0) ^a	28.4 (± 6.9) ^a					
Very dissatisfied	1	26.0 ^a	8.0 ^a	12.0 ^a	25.0 ^a					
Not applicable	44	18.9 (± 6.0) ^a	5.6 (± 4.5) ^a	6.5 (± 4.8) ^a	27.7 (± 6.6) ^a					
Feel valued by the dairy industry						0.5	0.3	0.4	0.4	
Extremely valued	12	16.6 (± 6.0) ^a	4.50 (± 5.81) ^a	5.08 (± 6.52) ^a	31.2 (± 5.5) ^a					
Very valued	25	16.6 (± 6.3) ^a	4.28 (± 4.15) ^a	4.68 (± 4.19) ^a	28.8 (± 4.5) ^a					
Moderately valued	43	16.6 (± 6.0) ^a	4.53 (± 3.55) ^a	4.63 (± 3.77) ^a	27.7 (± 6.4) ^a					
Slightly valued	17	18.8 (± 8.8) ^a	4.88 (± 3.77) ^a	5.71 (± 3.75) ^a	26.1 (± 6.1) ^a					

Continued

Table 7 (Continued). Comparing social environment variables with mental health scores of 115 dairy farmers in Western Canada and Ontario using ANOVA and *t*-tests

Variable	n	Mental health outcomes (mean $\pm$ SD)					P-value		
		Stress (PSS) ¹	Anxiety (GAD-7) ²	Depression (PHQ-9) ³	Resilience (CD-RISC) ⁴	Stress	Anxiety	Depression	Resilience
Not at all valued	17	18.7 (± 7.6) ^a	7.00 (± 5.17) ^a	6.71 (± 4.69) ^a	28.0 (± 7.2) ^a	<0.001	<0.001	0.001	0.005
Not applicable	1	26.0 ^a	7.00 ^a	11.0 ^a	27.0 ^a				
Feel valued by consumers	2	16.0 (± 15.6) ^{bc}	10.5 (± 14.8) ^a	12.0 (± 17.0) ^a	37.5 (± 3.5) ^a				
Extremely valued	22	13.2 (± 6.0) ^c	2.18 (± 2.02) ^b	2.91 (± 2.71) ^b	30.6 (± 4.5) ^a				
Very valued	54	16.6 (± 6.0) ^{bc}	4.26 (± 3.77) ^{ab}	4.83 (± 4.04) ^{ab}	28.3 (± 6.3) ^{ab}				
Moderately valued	25	20.0 (± 6.4) ^{ab}	6.36 (± 3.81) ^a	6.36 (± 3.73) ^a	25.6 (± 5.8) ^b	—	—	—	—
Slightly valued	12	22.6 (± 6.1) ^a	8.92 (± 3.85) ^a	7.58 (± 4.29) ^a	26.1 (± 5.4) ^{ab}				
Not at all valued	0	—	—	—	—				
Not applicable	0	—	—	—	—				

^{a-c}Mean mental health values in the same column under the same table section with different superscripts are statistically different ($P < 0.05$) or tend to be significant ($P < 0.1$).

¹Perceived Stress Scale (Cohen et al., 1983), which contains ten questions relating to the respondents' feelings over the past month, scoring each question from 0 to 4 and summing them to get an overall score out of 40, where a greater score indicates a higher level of perceived stress.

²General Anxiety Disorder-7 (Spitzer et al., 2006), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 21, where a score of 10 or higher is considered a reasonable threshold for identifying cases of generalized anxiety disorder.

³Patient Health Questionnaire (Kroenke et al., 2001), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 27, where a score of 10 or higher is considered a reasonable threshold for cases of depression.

⁴Connor-Davidson Resilience Scale-10 (Connor and Davidson, 2003), using 10 questions to assess resilience, or the ability of a person to cope or thrive despite hardships, over the past month. Each question had a score of 0 to 4, and scores were summed to get an overall score of 40, where a greater score indicates greater resilience.

exhibited lower stress, anxiety, and depression scores than those who described their support as “dissatisfied” or “neutral” ($P < 0.05$; Table 7). These associations were accompanied by large effect sizes for stress ($f = 0.53$), anxiety ($f = 0.53$), and depression ($f = 0.41$), indicating substantial differences linked to perceived support quality. In addition, farmers who were very satisfied with professional work support demonstrated higher resilience compared with those who felt neutral toward their work support (30.4 ± 5.17 vs. 26.3 ± 6.4 ; $P = 0.04$). The associations between perceived work-support quality and mental health outcomes were characterized by effect sizes of moderate magnitude (Supplemental Table S6).

Research has focused on the interplay between social relationships and mental health outcomes in agricultural settings, particularly among dairy farmers. As Kawachi and Berkman (2001) noted, social connections can protect against workplace mental health challenges. Nevertheless, managing the demands of personal relationships, leisure activities, and self-care remains a challenge for many individuals, which can exacerbate mental health concerns.

These findings collectively highlight the substantial effect of familial relationships in dairy farming. They suggest that interactions with family members correlate with improved mental health outcomes, underscoring the necessity for fostering supportive environments at home and in the workplace to enhance farmers' well-being. Collaborating with family members presents advantages and challenges. The inherent trust within familial relationships often fosters open communication. Families typically share fundamental values and a strong sense of unity, which is crucial for preserving legacies and offering essential psychosocial support (Jabbari et al., 2023). Nevertheless, the need to maintain professional boundaries in a family-owned business should be recognized. In Canadian dairy operations, family ownership is prevalent, emphasizing the deep connection between family units and farming. The intricate relationships among the farm, family, and home create unique opportunities and challenges (McCurdy and Carroll, 2000). Collaborative efforts within this family setting promote a shared vision that integrates economic and social objectives, ultimately enhancing the farm's profitability. However, economic pursuits can be impeded by unresolved intergenerational tensions arising from differences in age, life experiences, belief systems, and individual needs. Such family discord can negatively affect the farm's financial health, particularly as conflicts intensify under stress, with farm life offering little separation from between personal and work-related matters (Finnigan, 2019; Mendly-Zambo, 2023). Recognizing these stressors can foster understanding and forgiveness among family members, promoting the maintenance of

positive dynamics. Despite the potential challenges, the family's and the farm's sustainability heavily depends on the commitment to nurturing and sustaining strong familial relationships (Jabbari et al., 2023). Conversely, hiring external employees introduces diverse perspectives, fosters professional growth, promotes meritocracy, and provides networking opportunities, but can also come with potential challenges, including a need for increased trust and greater flexibility regarding work hours. The findings also highlight the significance of social support, both at home and in the workplace, in contributing to farmers' mental well-being.

Looking at physical health, the number of hours of sleep was associated with who the farmers worked with ($P = 0.004$) and whether they had someone to confide in ($P < 0.001$). These variables demonstrated moderate associations, with Cramer's V values of 0.3 and 0.4, respectively. A greater proportion of farmers who work with family members and employees, and those who work solely with family, tend to get more sleep (Table 8). This suggests that managing employees may not only contribute to anxiety or depression but could also consume time that would otherwise be available for sleep. Because these are univariable analyses, more work should assess the multitude of factors associated with sleep and the social environment on the farm.

Regarding professional relationships, the study indicates that many farmers express high levels of satisfaction with the support received from their veterinarians and feed representatives. Specifically, 52% of farmers ($n = 60$) reported being "highly satisfied" with their veterinarian, and 50% ($n = 57$) reported similar satisfaction with their feed representative. Additionally, 30% ($n = 34$) of farmers were "satisfied" with their veterinarians and 40% ($n = 46$) with their feed representatives. The percentage of farmers who felt "neutral" about their veterinarian and feed representative was low at 3% ($n = 3$) and 2% ($n = 2$), respectively, highlighting their overall positive experiences. Dissatisfaction was minimal, with only 0.87% of farmers voicing discontent with their veterinarian and none reporting dissatisfaction with their feed representative. This difference could be attributed to the relative ease of switching feed companies compared with the challenge of finding a new veterinarian. When comparing the relationship between professional satisfaction and mental health outcomes, notable differences emerged in farmers' satisfaction levels regarding their feed representatives and equipment suppliers. Farmers who expressed higher satisfaction reported lower depression scores compared with those who either lacked support or were dissatisfied with these professionals (Table 7). The findings emphasize the crucial role of social connections and strong support systems within the farming community, indicating that these relation-

ships can significantly improve mental health outcomes for farmers. It is essential for farmers to prioritize effective feeding methods and maintain high standards of cow health because these factors directly influence the quality of milk produced. By working closely with reliable veterinarians and experienced feed representatives, farmers can alleviate operational challenges and ensure their livestock are cared for, from both health and nutrition perspectives. Such collaborations not only enhance productivity but also foster a sense of community and mutual support, ultimately contributing to better mental well-being among farmers.

The results revealed that 32% of farmers felt extremely or very valued by the dairy industry, whereas 24% felt similarly by consumers, contrasting with 15% and 10% of participants who did not feel valued by the industry and consumers, respectively. Although feelings of being valued by the dairy industry were not associated with mental health outcomes, there were significant associations with large effect sizes between how valued farmers felt by consumers and their stress, anxiety, depression, and resilience scores (Supplemental Table S6). Notably, those farmers who felt highly valued by consumers had significantly lower stress scores (13.2 ± 6.0) compared with their peers who felt undervalued (22.6 ± 6.1 ; $P < 0.001$). Additionally, lower anxiety scores were reported by those feeling highly valued (2.18 ± 2.02) in contrast to both those who felt not valued (8.92 ± 3.85 ; $P < 0.001$) and those who only felt slightly valued (6.31 ± 3.81 ; $P = 0.002$). Similar associations were noted concerning depression, with highly valued farmers reporting lower scores (3.84 ± 4.29) than those not feeling valued (6.36 ± 3.73 ; $P = 0.04$). In terms of resilience, farmers who felt slightly valued demonstrated lower resilience (25.6 ± 5.76) compared with those who felt very valued (30.5 ± 5.95 ; $P = 0.03$) and extremely valued (37.5 ± 3.5 ; $P = 0.05$). No associations and small effect sizes were observed between physical health outcomes and the support received at home or at work, the support provided by agricultural professionals, or the feeling of being valued by consumers or the dairy industry ($P > 0.1$). These results suggest that a significant relationship may exist between farmers' perceptions of support and their mental health outcomes, highlighting how recognition from the public can affect their well-being, or perhaps another related or confounding variable was involved. This validation of farmers' contributions through appreciation recognizes their labor and reinforces their role in society, offering greater meaning and purpose to their work. The disparity between public perception of farming and the lived realities of agriculture has been well-documented, and addressing this gap could lead to improved mental health outcomes for farmers (Harandi et al., 2017; Finnigan, 2019).

Table 8. Comparing social environment variables with sleep outcomes of 115 dairy farmers in Western Canada and Ontario using chi-squared analyses

Variable	Physical health outcomes					P-value	
	Getting enough sleep		Sleeping hours			Getting enough sleep	Sleeping hours
	No	Yes	7–8 h	5–6 h	<5 h		
Total workers ($\pm$ SD)	8.48 ($\pm$ 15.7)	4.37 ($\pm$ 3.46)	5.08 ($\pm$ 6.34)	8.11 ($\pm$ 15.4)	3.83 ($\pm$ 2.13)	0.002	0.9
Who works on the farm (%)						0.5	0.004
Employees	3	1	3	1	0		
Family members	16	17	13	20	1		
Family members and employees	35	26	25	32	4		
Someone to talk to at work (%)						0.1	<0.001
Yes	43	40	9	47	3		
No	12	4	32	6	2		
Blank	1		0	0	1		
Satisfied with support at home (%)						0.1	0.6
Very satisfied	12	18	11	16	3		
Satisfied	22	15	12	24	0		
Neutral	13	9	13	7	2		
Dissatisfied	3	3	3	3	0		
Very dissatisfied	4	0	2	2	1		
Not applicable	1	0	0	0	1		

Housing and Animal Management

The average number of cows managed by participating farmers was 150 ± 160 , ranging from 30 to 1,100. The study average aligns closely with the Western Canadian average of 152 ± 22 ($P = 0.3$), as reported by the Dairy Farmers of Canada (2021). However, it differed significantly from the averages recorded in Quebec and Ontario, which stood at 85 ± 10 ($P < 0.001$). Most participants used freestalls for lactating cows (79%, $n = 91$). Tiestalls were used by 14% ($n = 16$), and 7% ($n = 8$) used bedded packs. Regarding milking methods, 59% used conventional pipelines ($n = 68$), and 41% used automated milking systems (AMS; $n = 47$). More detailed characteristics of the housing and animal management variables are presented in Table 9.

Statistical analysis revealed no significant associations of mental health scores related to the number of cows, milking systems, or secondary housing bases ($P > 0.1$; Table 9). Effect sizes for these comparisons were small, indicating minimal practical differences across these variables (Supplemental Table S6). However, a notable difference in depression scores was found based on housing type, with a moderate effect size ($P = 0.002$; $f = 0.34$); farmers using bedded-pack systems reported the highest average depression score of 10.1 ± 7.1 , those using freestall systems had a lower average score of 4.69 ± 3.80 ($P = 0.002$), and farmers with tiestall systems recorded an average of 5.69 ± 4.38 ($P = 0.04$). Regarding stall base, there was also a tendency for farmers with deep-bedded stalls to experience higher anxiety ($P = 0.09$) and depression scores ($P = 0.06$) compared with those using concrete as a stall base. Moreover, depres-

sion scores tended to be associated with bedding type ($P = 0.06$); farmers using a mix of organic and inorganic bedding showed higher depression scores compared with those who used only organic bedding. Stall base and bedding type had a moderate effect size for anxiety and depression scores (Supplemental Table S6).

The cost of bedding and the availability of bedding materials are primary issues for dairy farmers (Barberg et al., 2007; Endres, 2009). Annually, bedded packs and similar deep bedding systems rank as the costliest bedding options per cow (Smith et al., 2017). Additionally, Endres (2012) found that the bedding expenses for farmers using bedded packs were notably higher compared with those using freestalls. This disparity in cost could potentially cause financial stress, negatively affecting the mental health of farmers. However, bedded packs are also known to lower the incidence of lameness and improve cow comfort (Lobeck et al., 2011), which could reduce veterinary bills and labor needs.

Regarding sleep, there were no associations of sleep variables with the state of the barn, housing, stall base, and bedding (Supplemental Table S3, see Notes). Milking system tended to be associated with the amount of sleep reported by participants ($P = 0.07$); a higher percentage of farmers using conventional milking systems (36%) reported getting 5 to 6 h of sleep per night compared with only 16% of those using AMS, and a greater proportion (22%) of AMS users reported sleeping between 7 and 8 h compared with 19% of conventional users. Both groups had a similar proportion (3.5%) of members reporting less than 5 h of sleep per night (Supplemental Table S3). Furthermore, regarding sleep adequacy, a higher percentage of farmers using conventional milking systems

Table 9. Comparing housing and animal management variables with mental health scores of 115 dairy farmers in Western Canada and Ontario using ANOVA and *t*-tests

Variable	n	Mental health outcomes (mean $\pm$ SD)				P-value			
		Stress (PSS) ¹	Anxiety (GAD-7) ²	Depression (PHQ-9) ³	Resilience (CD-RISC) ⁴	Stress	Anxiety	Depression	Resilience
Milking system						0.3	0.8	0.8	0.9
AMS	47	16.5 (± 7.3) ^a	5.02 (± 3.82) ^a	5.06 (± 3.93) ^a	28.1 (± 6.4) ^a				
Conventional	68	17.8 (± 6.3) ^a	4.84 (± 4.56) ^a	5.31 (± 4.65) ^a	28.1 (± 5.8) ^a				
State of the barn						0.04	0.5	0.7	0.3
New construction	34	15.0 (± 7.1) ^b	4.68 (± 4.76) ^a	5.41 (± 5.26) ^a	29.6 (± 5.3) ^a				
Original construction	52	18.1 (± 6.6) ^{ab}	5.06 (± 3.86) ^a	5.12 (± 4.09) ^a	27.7 (± 4.1) ^a				
Retrofit	18	16.9 (± 6.8) ^{ab}	4.00 (± 3.34) ^a	4.39 (± 4.07) ^a	26.9 (± 4.1) ^a				
Other	11	21.1 (± 3.8) ^a	6.45 (± 5.68) ^a	6.36 (± 2.84) ^a	27.0 (± 2.8) ^a				
Housing						0.5	0.1	0.002	0.5
Bedded pack	8	20.0 (± 5.4) ^a	7.88 (± 6.40) ^a	10.1 (± 7.1) ^a	28.9 (± 5.4) ^a				
Freestall	91	17.1 (± 6.8) ^a	4.67 (± 4.06) ^a	4.69 (± 3.80) ^b	28.3 (± 6.2) ^a				
Tiestall	16	17.4 (± 7.3) ^a	4.81 (± 3.80) ^a	5.69 (± 4.38) ^b	26.5 (± 5.6) ^a				
Stall base						0.3	0.09	0.06	0.8
Concrete	76	16.9 (± 7.1) ^a	4.41 (± 4.04) ^b	4.66 (± 4.24) ^b	28.2 (± 4.2) ^a				
Deep bedded	39	18.2 (± 5.9) ^a	5.90 (± 4.55) ^a	6.28 (± 4.41) ^a	27.9 (± 4.4) ^a				
Secondary base						0.6	0.4	0.3	0.9
Compost	6	18.5 (± 4.7) ^a	5.50 (± 2.74) ^a	8.67 (± 3.73) ^a	28.2 (± 5.0) ^a				
Mattress	49	17.0 (± 7.4) ^a	4.76 (± 4.38) ^a	5.16 (± 4.87) ^a	28.3 (± 5.5) ^a				
Mattress + water beds	6	16.5 (± 7.1) ^a	4.00 (± 4.90) ^a	4.50 (± 3.33) ^a	28.8 (± 7.9) ^a				
Sand	32	18.0 (± 6.0) ^a	5.94 (± 4.80) ^a	5.78 (± 4.29) ^a	28.2 (± 7.0) ^a				
Straw or sawdust	10	15.5 (± 6.2) ^a	3.70 (± 2.63) ^a	3.50 (± 3.80) ^a	26.9 (± 7.1) ^a				
Water beds	11	17.0 (± 7.3) ^a	3.45 (± 3.24) ^a	3.73 (± 2.37) ^a	27.4 (± 5.0) ^a				
Bedding						0.7	0.1	0.06	0.9
Inorganic	27	17.8 (± 6.1) ^a	5.78 (± 5.11) ^a	5.26 (± 4.23) ^{ab}	28.6 (± 7.3) ^b				
Organic	85	17.0 (± 6.9) ^a	4.51 (± 3.94) ^a	4.99 (± 4.33) ^b	27.9 (± 5.6) ^b				
Organic + inorganic	3	20.0 (± 7.9) ^a	8.67 (± 1.53) ^a	11.0 (± 2.6) ^a	28.0 (± 6.6) ^a				

^{a,b}Mean mental health values in the same column under the same table section with different superscripts are statistically different ($P < 0.05$) or tend to be significant ($P < 0.1$).

¹Perceived Stress Scale (Cohen et al., 1983), which contains ten questions relating to the respondents' feelings over the past month, scoring each question from 0 to 4 and summing them to get an overall score out of 40, where a greater score indicates a higher level of perceived stress.

²General Anxiety Disorder-7 (Spitzer et al., 2006), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 21, where a score of 10 or higher is considered a reasonable threshold for identifying cases of generalized anxiety disorder.

³Patient Health Questionnaire-9 (Kroenke et al., 2001), which contains 7 questions used to score respondents' feelings over the past week. Responses were scored from 0 to 3; the maximum score for each was out of 27, where a score of 10 or higher is considered a reasonable threshold for cases of depression.

⁴Connor-Davidson Resilience Scale-10 (Connor and Davidson, 2003), using 10 questions to assess resilience, or the ability of a person to cope or thrive despite hardships, over the past month. Each question had a score of 0 to 4, and scores were summed to get an overall score of 40, where a greater score indicates greater resilience.

felt they needed more sleep than their counterparts with AMS (36% vs. 19%). When examining the association between milking systems and sleep duration, a moderate effect size was observed (Cramer's $V = 0.31$), indicating a potential relationship that may warrant further investigation in studies with larger sample sizes.

The study found that technology is being used in various aspects of farming. According to the survey, 77% of farmers ($n = 89$) reported using activity monitors. In contrast, only 13% of farmers ($n = 15$) indicated using automated feeding systems. Additionally, 58% of farmers ($n = 67$) use automated manure scrapers, 45% ($n = 52$) use milk quality monitors, and 53% ($n = 61$) employ rumination monitors. However, we found no associations between well-being outcomes and technology use ($P > 0.1$ and $f < 0.5$). Farmers using newly constructed

barns reported lower stress scores compared with those working in barns that combined new, retrofitted, and older structures, while farms with exclusively original or retrofitted barns showed intermediate stress levels (Table 9). The overall association demonstrated a moderate effect size ($f = 0.30$).

The assessment of the psychological well-being of dairy farmers in relation to animal management practices requires further exploration. The increase in the scale of dairy operations often correlates with heightened revenue, additional employment opportunities, the financial capacity to integrate more technology to alleviate labor burdens, and an expansion of farm responsibilities. These changes can also lead to elevated costs and potential disruptions in work-life balance. A consistent shortage of skilled farm labor often compels dairy operations to hire

foreign workers to fill the gap, but this reliance introduces challenges, including language barriers that may impede effective communication, potentially resulting in mismanagement of animal care, including issues related to disease detection and treatment (Barkema et al., 2015). Such mismanagement could lead to diminished care for the animals and increased veterinary expenses. Moreover, the financial challenges associated with agricultural work are underscored by the theme of inadequate wages, as highlighted in Mendlly-Zambo's (2023) report, where one farmer noted the near impossibility of sustaining a living as a farmer without exploiting either the environment or laborers. The expansion of the farm can exacerbate difficulties in labor recruitment and retention because increased scale brings added pressures on both farmers and their workforce. Interestingly, the current study revealed no correlation between farmers' mental health scores and farm size, as measured by the number of cows. One possible explanation for this finding is that farmers, whether managing smaller or larger operations, may have become accustomed to their workloads over time, thereby reducing the stress associated with such responsibilities. Furthermore, the relationship between farmer well-being and herd size may not be linear but better understood within a multidimensional framework that considers various factors, such as the number of employees, veterinary expenses, and management practices.

The effective implementation of management practices, combined with the adoption of technological innovations, can enhance milk production while promoting better health and welfare for dairy cows (Jacobs and Siegford, 2012; Tse et al., 2018; Ventura et al., 2021). King et al. (2021) posited that using AMS could potentially lead to better mental health for farmers; however, our current study did not yield similar results within our sample. Nevertheless, a Canadian investigation by Tse et al. (2018) highlighted that dairy farmers employing AMS perceived their work as less stressful and physically demanding. Our findings provide partial support for this claim: Individuals using AMS reported better sleep quality compared with their conventional counterparts. The primary benefit of automation lies in its capacity to substitute manual labor during milking (Acemoglu and Restrepo, 2019), although it does not inherently reduce the total number of working hours but rather alters the nature and flexibility of labor requirements. A reduction in physical discomfort, coupled with enhanced scheduling flexibility, may enable farmers to concentrate their energy and resources on other farm tasks. Ultimately, AMS offer farmers a more manageable lifestyle by alleviating the necessity of being present during the milking process, which traditionally occupies considerable time (Tse et al., 2018). It is important to note that phenomena such as postpurchase rationalization, where individuals

downplay the shortcomings of products to validate their choices (Chatzidakis et al., 2009), and choice-supportive bias, attributing favorable characteristics to selected options while viewing alternatives negatively (Kafaei et al., 2021), could also be at play.

Overall, the absence of associations between mental health scores and milking system, technology use, and many housing characteristics suggests that farmers can operate proficiently in a variety of production systems. It is also possible that our limited sample of farmers was not sufficient to detect differences across these variables.

Financial and Transition Planning

The survey results showed that 31% of the surveyed participants reported having a documented business plan, indicating that most businesses lacked this crucial strategic tool. Similarly, we found that 36% of the participants had a written succession plan, highlighting a potential gap in long-term planning to ensure the continuity of their businesses. These proportions are higher than the reported value by Statistics Canada (2022), which indicates that the percentage of farms with a succession plan increased from 8.4% to 12% between 2016 and 2021. Additionally, 18% of respondents reported conducting annual risk assessments while maintaining written management or contingency plans. Furthermore, findings from another Canadian study echoed these observations, revealing that many farmers relied on informal plans and initiated succession planning only when a family member expressed interest in pursuing a career in dairy farming (Finnigan, 2019). Although financial and transition-planning variables were not significantly associated with farmer well-being ($P > 0.1$) and demonstrated only small effect sizes, these patterns may nonetheless point to important underlying dynamics. The absence of strong statistical relationships could indicate that current planning practices are insufficiently developed to meaningfully influence well-being or that there are not enough farmers with written plans to detect differences. Together, these findings highlight a potential need for more proactive, structured, and accessible planning resources within the agricultural community.

Specific Stressors

Participants in the study identified their greatest sources of stress in dairy farming as workload (47%), feed prices (43%), farm finances (42%), and weather (41%; Supplemental Figure S1, see Notes). Interestingly, our farmers identified flooding (85%), wildfires (74%), and learning and implementing new technology (62%) as the least stress-inducing factors. There were positive associations with large effect sizes between anxiety and depres-

sion scores and concerns about wildfires ($P < 0.001$ and $P = 0.001$, respectively; Supplemental Tables S4, S6) as well as concerns about flooding ($P = 0.02$ and $P < 0.001$, respectively). There were no significant differences in mental health scores when comparing drought concerns, pasture conditions, weather, or quota values. However, several of these variables showed moderate effect sizes in relation to mental health outcomes, suggesting that the study may have been underpowered to detect significant differences. As such, these results remain inconclusive but point toward the possibility of meaningful underlying associations that could be clarified in a larger or more targeted sample (Supplemental Table S6).

Significant differences emerged when examining mental health scores in relation to animal health and welfare concerns. For example, farmers who were very often concerned about animal health showed higher anxiety (8.71 ± 6.07 vs. 3.15 ± 3.54 ; $P = 0.02$) and depression scores (9.86 ± 7.77 vs. 3.65 ± 3.26 ; $P = 0.01$) compared with those who were never concerned. Further exploration of the magnitude of these effects revealed large effect sizes for the association between animal health and welfare concerns and stress ($f = 0.40$), anxiety ($f = 0.50$), and depression scores ($f = 0.42$). Further, stress, anxiety, and depression were found to be significantly associated with concerns about farm finances. Farmers who frequently worried about their farm finances experienced higher scores for stress (21.5 ± 6.24 vs. 13.1 ± 7.2 ; $P < 0.001$; $f = 0.51$), anxiety (8.25 ± 5.18 vs. 2.25 ± 3 ; $P < 0.001$; $f = 0.57$), and depression (8.68 ± 4.81 vs. 3.06 ± 3.77 ; $P < 0.001$; $f = 0.54$) compared with those who were not concerned about farm finances. In addition, farmers who were frequently worried about work load experienced higher stress (22.8 ± 5.4 vs. 11.3 ± 5.2 ; $P < 0.001$; $f = 0.8$), anxiety (7.72 ± 4.74 vs. 1.40 ± 1.77 ; $P < 0.001$; $f = 0.6$), and depression (8.12 ± 4.88 vs. 2.50 ± 2.63 ; $P = 0.004$; $f = 0.6$), as well as lower resilience (25.6 ± 5.9 vs. 32.1 ± 6.6 ; $P = 0.05$; $f = 0.43$) compared with those who were never worried about work load. Similar patterns were observed for other concerns such as energy pricing, feed availability, feed cost, human resource management, learning new technology, personal finances, policy and regulations, public perceptions, continuation of quota, raw milk prices, marketing milk prices, and animal rights activists (Supplemental Tables S4 and S6, see Notes).

Notable relationships were observed between the number of hours slept and various concerns affecting farmers' lives. Farmers who experienced less concern about their personal finances ($P = 0.02$), the management of human resources ($P < 0.001$), and their overall workload ($P = 0.002$) reported getting more sleep. Conversely, those who were increasingly concerned about the welfare of their animals tended to get less sleep. This pattern suggests that farmers may experience a strong sense of obli-

gation to work hard and continually maintain the health of their cows, potentially sacrificing their own sleep or finding it difficult to sleep due to anxiety.

These findings were consistent with previous studies, which also identified unpredictable weather and economic pressures as the main stressors for farmers (Finnigan, 2019; Jones-Bitton et al., 2020). Additionally, the Canadian Agricultural Safety Association (2005) reported that poor production and farm finances were the top stressors in farming. Another study by Sturgeon and Morrisette (2010) found that financial strain and a lack of control over farming variables such as weather and farm culture were major stressors in this sector. Although climate change and extreme weather events, such as drought, flooding, and wildfires, have been recognized as significant sources of financial stress for the dairy industry (Abbass et al., 2022; Rocha et al., 2022), these were not the key stressors reported in the current study. Witnesses in the Finnigan (2019) report mentioned that farmers have been accustomed to dealing with weather uncertainties for generations. Nevertheless, weather remains an uncertainty that can cause stress, with those participants most concerned about the weather experiencing poorer mental well-being. Finally, many farmers did not report feeling overly concerned about learning and adopting new technological advances, despite the potential challenges and overwhelming amount of data collected (Jacobs and Siegford, 2012).

Study Limitations and Future Directions

Because participation in this study was entirely voluntary, response bias is possible. Recruitment through email and social media could have disproportionately reached younger or more progressive, technologically engaged farmers, while excluding those with limited internet access, lower digital literacy, or infrequent social media use, groups that often include older farmers or those residing in remote or rural areas. This bias may partially account for the relatively high proportion of AMS users and younger respondents in our sample, which limits the generalizability of the findings to the broader dairy farming population. For instance, 41% of farmers in our study reported using AMS, compared with a published estimate of 22% for farms in the same provinces during the study year (Government of Canada, 2023); although their estimate likely underrepresents the true prevalence of AMS herds due to lower participation in milk-recording programs, the discrepancy nonetheless suggests that more progressive producers may have been more inclined to participate, thereby influencing estimates of management practices and mental health outcomes.

Nonresponse bias may have also occurred if certain farmers were reluctant to participate due to poor mental

health, burnout, stigma, privacy concerns, or discomfort sharing personal information. Previous research indicates that individuals experiencing poor mental health tend not to engage in surveys, which could result in an underreporting of mental health issues (Torvik et al., 2012). As a result, our sample may disproportionately reflect farmers who felt more comfortable discussing mental health, who were experiencing better mental health, or who had sufficient time and energy to participate, rather than those experiencing the most severe strain. Alternatively, we are also aware of farmers who participated because of their experiences with high stress or mental health challenges and eagerness to support our research. Regarding the timing of survey completion, most participants completed the survey during late spring, summer, and early fall—periods that coincide with high stress and potentially leading our estimates to be skewed or reflective of particularly busy times.

Further, not all survey components used validated scales, such as assessment of physical health and sleep, work-life balance, and social support variables, which were measured using study-specific questions rather than standardized scales. This distinction is important for interpreting the findings and understanding the limitations of these specific measures.

Because our study was exploratory and hypothesis-generating, we used inferential statistical tests not to test predefined hypotheses, but rather to identify preliminary patterns and potential relationships across variables that can inform future, more targeted investigations. In this context and our limited sample, the significant associations identified reveal possible, but not all, factors associated with dairy farmer well-being. Although several variables in this study did not reach significance based on their *P*-values, many nonetheless demonstrated moderate-to-large effect sizes, indicating that these environmental and operational stressors may meaningfully influence mental health outcomes despite the lack of statistically detectable differences. Effect sizes of this magnitude typically reflect nontrivial practical differences, and when paired with nonsignificant results, they often suggest insufficient statistical power, substantial variability within groups, or both. Taken together, these findings point to potentially important relationships that merit further investigation. Future studies with larger sample sizes or more refined measures will be essential to determine whether these factors exert substantive effects on farmer mental health.

A final limitation of this study is the reliance on univariate analyses. Although the authors intentionally chose to avoid multivariable modeling, the lack of adjustment for significant confounding factors—including age, gender, socioeconomic status, lifestyle behaviors, and pre-existing health conditions—limits the interpret-

ability of our findings. Therefore, our work highlights a potential gap in traditional analytical methods employed in the study of complex systems; thus, our accompanying manuscript (Tambadou et al., 2026) investigates dairy farmer well-being using a multifactorial framework.

CONCLUSIONS

From a sample of 115 dairy farmers in Western Canada and Ontario, we found several univariable associations between work-life balance, social environments, and support systems, as well as specific stressors unique to dairy farming. Factors associated with better mental well-being were generally related to participants' social environment, such as feeling supported at home, being able to talk to someone at work, and working with family members as opposed to nonfamily employees. Further, those who felt highly valued by consumers had lower stress, anxiety, and depression scores and higher resilience scores than those who felt less valued. The main stressors participants identified were financial issues, feeding costs, weather variability, and their overall workload. In general, those who were more concerned by such stressors had poorer mental well-being scores.

NOTES

This study was financially supported by BC Dairy (Langley, BC, Canada), Alberta Milk (Edmonton, AB, Canada), SaskMilk (Regina, SK, Canada), Dairy Farmers of Manitoba (DFM; Winnipeg, MB, Canada), Dairy Farmers of Ontario (DFO; Mississauga, ON, Canada), and Mitacs (Vancouver, BC, Canada). We are grateful to the participating dairy farmers who have been affected by the topic of this study and who graciously lent their time. Supplemental material for this article is available at <https://doi.org/10.5683/SP3/CBZKRV>. Author contributions are as follows: H. Tambadou, conceptualization, data curation, formal analysis, methodology, and writing (original draft, review and editing); B. Zwick, conceptualization, data curation, methodology, and writing (review and editing); A. Le Heiget, conceptualization, data curation, methodology, and writing (review and editing); B. Hagen, conceptualization, formal analysis, funding acquisition, methodology, and writing (review and editing); C. Winder, conceptualization, formal analysis, funding acquisition, methodology, and writing (review and editing); K. Plaizier, conceptualization, formal analysis, methodology, and writing (review and editing); J. Kinley, conceptualization, funding acquisition, methodology, and writing (review and editing); K. Ominski, conceptualization, methodology, and writing (review and editing); E. Pajor, conceptualization, funding acquisition, and methodology; H. Watson, conceptualization, funding

acquisition, and methodology; M. King, conceptualization, data curation, funding acquisition, methodology, project administration, supervision, and writing (original draft, review and editing). The study protocol received approval from the Research Ethics Board at the University of Manitoba (HE2022-1459), and informed consent was obtained from all participants before starting the questionnaire and throughout the study. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AMS = automated milking system; CD-RISC = Connor–Davidson Resilience Scale-10; GAD-7 = Generalized Anxiety Disorder-7; MHRC = Mental Health Research Canada; PHQ-9 = Patient Health Questionnaire-9; PSS = Perceived Stress Scale.

REFERENCES

- Abbass, K., M. Z. Qasim, H. Song, M. Murshed, H. Mahmood, and I. Younis. 2022. A review of the global climate change impacts, adaptation, and sustainable mitigation measures. *Environ. Sci. Pollut. Res. Int.* 29:42539–42559. <https://doi.org/10.1007/s11356-022-19718-6>.
- Acemoglu, D., and P. Restrepo. 2019. Automation and new tasks: How technology displaces and reinstates labor. *J. Econ. Perspect.* 33:3–30. <https://doi.org/10.1257/jep.33.2.3>.
- Agriculture and Agri-Food Canada. 2025. Mental health in agriculture. Government of Canada. Accessed Dec. 28, 2025. <https://agriculture.canada.ca/en/departement/initiatives/mental-health>.
- Altevogt, B. M., H. R. Colten, and M. Bruce. 2006. Sleep Disorders and Sleep Deprivation: An Unmet Public Health Problem. National Academies Press.
- Barberg, A. E., M. I. Endres, and K. A. Janni. 2007. Dairy compost barns in Minnesota: A descriptive study. *Appl. Eng. Agric.* 23:231–238. <https://doi.org/10.13031/2013.22606>.
- Barkema, H. W., M. A. G. von Keyserlingk, J. P. Kastelic, T. J. G. M. Lam, C. Luby, J.-P. Roy, S. J. LeBlanc, G. P. Keefe, and D. F. Kelton. 2015. *Invited review*: Changes in the dairy industry affecting dairy cattle health and welfare. *J. Dairy Sci.* 98:7426–7445. <https://doi.org/10.3168/jds.2015-9377>.
- Brennan, M., T. Hennessy, D. Meredith, and E. Dillon. 2022. Weather, workload and money: Determining and evaluating sources of stress for farmers in Ireland. *J. Agromedicine* 27:132–142. <https://doi.org/10.1080/1059924X.2021.1988020>.
- Bruce, S. E., K. A. Yonkers, M. W. Otto, J. L. Eisen, R. B. Weisberg, M. Pagano, M. T. Shea, and M. B. Keller. 2005. Influence of psychiatric comorbidity on recovery and recurrence in generalized anxiety disorder, social phobia, and panic disorder: A 12-year prospective study. *Am. J. Psychiatry* 162:1179–1187. <https://doi.org/10.1176/appi.ajp.162.6.1179>.
- Buisseret, F., N. Draye, C. Di Santo, J. Pacewicz, J. Pannetier, F. Dierick, and F. Telliez. 2024. Occupational risk factors for musculoskeletal disorders among workers in dairy diversification. *Healthcare (Basel)* 12:178. <https://doi.org/10.3390/healthcare12020178>.
- Campbell-Sills, L., and M. B. Stein. 2007. Psychometric analysis and refinement of the Connor–Davidson Resilience Scale (CD-RISC): Validation of a 10-item measure of resilience. *J. Trauma. Stress* 20:1019–1028. <https://doi.org/10.1002/jts.20271>.
- Canadian Agricultural Safety Association. 2005. National stress and mental survey of Canadian farmers. Accessed May 24 2024. https://www.casa-acsa.ca/wp-content/uploads/2005-02_National_Stress_Survey_Results.pdf.
- Chatzidakis, A., A. P. Smith, and S. Hibbert. 2009. “Do I need it, do I, do I really need this?”: Exploring the role of rationalization in impulse buying episodes. *Adv. Consum. Res.* 36:248–253.
- Chengane, S., C. L. Beseler, E. G. Duysen, and R. H. Rautiainen. 2021. Occupational stress among farm and ranch operators in the midwestern United States. *BMC Public Health* 21:2076. <https://doi.org/10.1186/s12889-021-12053-4>.
- Cohen, J. 1988. *Statistical Power Analysis for the Behavioral Sciences*. 2nd ed. Lawrence Erlbaum Associates, Hillsdale, NJ.
- Cohen, S., T. Kamarck, and R. Mermelstein. 1983. A global measure of perceived stress. *J. Health Soc. Behav.* 24:385–396. <https://doi.org/10.2307/2136404>.
- Cohen, S., T. Kamarck, and R. Mermelstein. 1994. Perceived stress scale. Pages 1–2 in *Measuring Stress: A Guide for Health and Social Scientists*. S. Cohen, R. C. Kessler, and L. G. Underwood, ed. Oxford University Press.
- Connor, K. M., and J. R. Davidson. 2003. Development of a new resilience scale: The Connor–Davidson resilience scale (CD-RISC). *Depress. Anxiety* 18:76–82. <https://doi.org/10.1002/da.10113>.
- Contzen, S., and I. Häberli. 2021. Exploring dairy farmers’ quality of life perceptions—A Swiss case study. *J. Rural Stud.* 88:227–238. <https://doi.org/10.1016/j.jrurstud.2021.11.007>.
- Dairy Farmers of Canada. 2021. How many cows are on Canadian dairy farms? Accessed May 10, 2024. <https://dairyfarmersofcanada.ca/en/our-commitments/animal-care/how-many-cows-farms-sizes>.
- Dean, A. G., K. M. Sullivan, and M. M. Soe. 2013. OpenEpi: Open source epidemiologic statistics for public health. Version 3.01. OpenEpi. Accessed Oct. 11, 2024. <https://www.openepi.com/SampleSize/SSPropor.htm>.
- Deary, I. J., J. Willock, and M. McGregor. 1997. Stress in farming. *Stress Med.* 13:131–136. [https://doi.org/10.1002/\(SICI\)1099-1700\(199704\)13:2<131::AID-SMI727>3.0.CO;2-T](https://doi.org/10.1002/(SICI)1099-1700(199704)13:2<131::AID-SMI727>3.0.CO;2-T).
- Dimich-Ward, H., J. R. Guernsey, W. Pickett, D. Rennie, L. Hartling, and R. J. Brison. 2004. Gender differences in the occurrence of farm related injuries. *Occup. Environ. Med.* 61:52–56.
- Doupbrate, D. I., D. Gimeno, M. W. Nonnenmann, G. R. Hagevoort, C. Rosas-Goulart, and J. C. Rosecrance. 2014. Prevalence of work-related musculoskeletal symptoms among U.S. large-herd dairy parlor workers. *Am. J. Ind. Med.* 57:370–379. <https://doi.org/10.1002/ajim.22286>.
- Doupbrate, D. I., L. Stallones, C. Lunner Kolstrup, M. W. Nonnenmann, S. Pinzke, G. R. Hagevoort, P. Lundqvist, M. Jakob, H. Xiang, L. Xue, P. Jarvie, S. A. McCurdy, S. Reed, and T. Lower. 2013. Work-related injuries and fatalities on dairy farm operations—A global perspective. *J. Agromedicine* 18:256–264. <https://doi.org/10.1080/1059924X.2013.796904>.
- Elliott, K. C., J. M. Lincoln, M. A. Flynn, J. L. Levin, M. Smidt, J. Dzigan, and A. K. Ramos. 2022. Working hours, sleep, and fatigue in the agriculture, forestry, and fishing sector: A scoping review. *Am. J. Ind. Med.* 65:898–912. <https://doi.org/10.1002/ajim.23418>.
- Endres, M. I. 2009. Compost bedded pack barns—Can they work for you? Pages 271–279 in *Western Canadian Dairy Seminar Advances in Dairy Technology*, Red Deer, AB, Canada. University of Alberta, Edmonton, AB, Canada. Accessed May 10, 2024. https://wcds.ualberta.ca/wcds/wp-content/uploads/sites/57/wcds_archive/Archive/2009/Manuscripts/CompostBedded.pdf.
- Endres, M. I. 2012. Bedding options for dairy cows. Pages 361–369 in *Western Canadian Dairy Seminar Advances in Dairy Technology*, Red Deer, AB, Canada. University of Alberta, Edmonton, AB, Canada. Accessed May 10, 2024. https://wcds.ualberta.ca/wcds/wp-content/uploads/sites/57/wcds_archive/Archive/2012/Manuscripts/Endres.pdf.
- Etherton, J. R., J. R. Myers, R. C. Jensen, J. C. Russell, and R. W. Braddee. 1991. Agricultural machine-related deaths. *Am. J. Public Health* 81:766–768. <https://doi.org/10.2105/AJPH.81.6.766>.
- Finnigan, P. 2019. Mental health: A priority for our farmers. Report of the Standing Committee on Agriculture and Agri-Food, 42nd Parliament, 1st Session. Speaker of the House of Commons, Ottawa, ON, Canada. <https://www.ourcommons.ca/Content/Committee/421/AGRI/Reports/RP10508975/agrip16/agrip16-e.pdf>.
- Fraser, C. E., K. B. Smith, F. Judd, J. S. Humphreys, L. J. Fragar, and A. Henderson. 2005. Farming and mental health problems and mental illness. *Int. J. Soc. Psychiatry* 51:340–349. <https://doi.org/10.1177/0020764005060844>.

- Government of Canada. 2022. Fact sheets. About the Canadian Dairy Commission. Accessed Nov. 20, 2023. <https://cdc-ccl.ca/en/node/888>.
- Government of Canada. 2023. Dairy barns by type. Accessed Nov. 20, 2023. <https://agriculture.canada.ca/en/sector/animal-industry/canadian-dairy-information-centre/statistics-market-information/farm-statistics/barns-type>.
- Hagel, L. M., J. A. Dosman, D. C. Rennie, M. W. Ingram, and A. Sent-hilselvan. 2004. Effect of age on hospitalized machine-related farm injuries among the Saskatchewan farm population. *J. Agric. Saf. Health* 10:155–162. <https://doi.org/10.13031/2013.16472>.
- Hagen, B. N. M., A. Sawatzky, S. L. Harper, T. L. O'Sullivan, and A. Jones-Bitton. 2021. What impacts perceived stress among Canadian farmers? A mixed-methods analysis. *Int. J. Environ. Res. Public Health* 18:7366. <https://doi.org/10.3390/ijerph18147366>.
- Hale, L. 2005. Who has time to sleep? *J. Public Health (Oxf.)* 27:205–211. <https://doi.org/10.1093/pubmed/fdi004>.
- Harandi, T. F., M. M. Taghinasab, and T. D. Nayeri. 2017. The correlation of social support with mental health: A meta-analysis. *Electron. Physician* 9:5212–5222. <https://doi.org/10.19082/5212>.
- Hassard, J., and K. Teoh. 2013. Psychosocial issues in the agriculture sector. Accessed May 14, 2024. <https://oshwiki.osha.europa.eu/en/themes/psychosocial-issues-agriculture-sector>.
- Heaton, K., A. Azuero, and D. Reed. 2010. Obstructive sleep apnea indicators and injury in older farmers. *J. Agromedicine* 15:148–156. <https://doi.org/10.1080/10599241003636020>.
- Hounsborne, B., R. T. Edwards, N. Hounsborne, and G. Edwards-Jones. 2012. Psychological morbidity of farmers and non-farming population: Results from a UK survey. *Community Ment. Health J.* 48:503–510. <https://doi.org/10.1007/s10597-011-9415-8>.
- Jabbari, B., C. Schoo, and A. S. Rouster. 2023. Family dynamics. In: *Stat-Pearls* [Internet]. Treasure Island, FL. Accessed May 10, 2024. <https://www.ncbi.nlm.nih.gov/books/NBK560487/?report=printable>.
- Jacobs, J. A., and J. M. Siegford. 2012. *Invited review*: The impact of automatic milking systems on dairy cow management, behavior, health, and welfare. *J. Dairy Sci.* 95:2227–2247. <https://doi.org/10.3168/jds.2011-4943>.
- Janssen, I., R. L. Berg, B. Marlena, and W. Pickett. 2019. Sleep in farm adolescents. *J. Rural Health* 35:436–441. <https://doi.org/10.1111/jrh.12341>.
- Johnson, A., L. Baccaglini, G. R. Haynatzki, C. Achutan, D. Loomis, and R. H. Rautiainen. 2021. Agricultural injuries among farmers and ranchers in the central United States during 2011–2015. *J. Agromedicine* 26:62–72. <https://doi.org/10.1080/1059924X.2020.1845268>.
- Jones-Bitton, A. 2019. Mental health in the dairy and livestock production sector. Pages 65–79 in *Western Canadian Dairy Seminar Advances in Dairy Technology*, Red Deer, AB, Canada. University of Alberta, Edmonton, AB, Canada. <https://wcds.ualberta.ca/wp-content/uploads/sites/57/2019/05/p-065-076-Jones-Bitton-WCDs-2019-Mental-Health-in-the-Dairy-and-Livestock-Production-Sector.pdf>.
- Jones-Bitton, A., C. Best, J. MacTavish, S. Fleming, and S. Hoy. 2020. Stress, anxiety, depression, and resilience in Canadian farmers. *Soc. Psychiatry Psychiatr. Epidemiol.* 55:229–236. <https://doi.org/10.1007/s00127-019-01738-2>.
- Kafaei, M., H. Marhamati, and S. Gharibzadeh. 2021. “Choice-supportive bias” in science: Explanation and mitigation. *Account. Res.* 28:528–543. <https://doi.org/10.1080/08989621.2021.1872377>.
- Kato, H., H. Ono, M. Sato, M. Noguchi, and K. Kobayashi. 2022. Relationships between management factors in dairy production systems and mental health of farm managers in Japan. *J. Dairy Sci.* 105:441–452. <https://doi.org/10.3168/jds.2021-20666>.
- Kawachi, I., and L. F. Berkman. 2001. Social ties and mental health. *J. Urban Health* 78:458–467. <https://doi.org/10.1093/jurban/78.3.458>.
- Kim, J. H. 2023. Cross-sectional study: The role of observation in epidemiological studies. Pages 19–42 in *Statistical Approaches for Epidemiology: From Concept to Application*. Springer International Publishing.
- King, M. T. M., R. D. Matson, and T. J. DeVries. 2021. Connecting farmer mental health with cow health and welfare on dairy farms using robotic milking systems. *Anim. Welf.* 30:25–38. <https://doi.org/10.7120/09627286.30.1.025>.
- Kivimäki, M., M. Jokela, S. T. Nyberg, A. Singh-Manoux, E. I. Fransson, L. Alfredsson, J. B. Bjorner, M. Borritz, H. Burr, A. Casini, E. Clays, D. De Bacquer, N. Dragano, R. Erbel, G. A. Geuskens, M. Hamer, W. E. Hoofman, I. L. Houtman, and K.-H. Jöckel. IPD-Work Consortium. 2015. Long working hours and risk of coronary heart disease and stroke: A systematic review and meta-analysis of published and unpublished data for 603 838 individuals. *Lancet* 386:1739–1746. [https://doi.org/10.1016/S0140-6736\(15\)60295-1](https://doi.org/10.1016/S0140-6736(15)60295-1).
- Kroenke, K., R. L. Spitzer, and J. B. Williams. 2001. The PHQ-9: Validity of a brief depression severity measure. *J. Gen. Intern. Med.* 16:606–613. <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>.
- Kurihara, S., and S. Shimoura. 2008. Quality control efforts among Japanese dairy farmers: Findings from National Field Surveys. *Hortic. Res.* 62:77–83.
- Lewis, M. Q., N. L. Sprince, L. F. Burmeister, P. S. Whitten, J. C. Torner, and C. Zwerling. 1998. Work-related injuries among Iowa farm operators: An analysis of the Iowa Farm Family Health and Hazard Surveillance Project. *Am. J. Ind. Med.* 33:510–517. [https://doi.org/10.1002/\(SICI\)1097-0274\(199805\)33:5<510::AID-AJIM11>3.0.CO;2-0](https://doi.org/10.1002/(SICI)1097-0274(199805)33:5<510::AID-AJIM11>3.0.CO;2-0).
- Lindahl, C., P. Lundqvist, G. R. Hagevoort, C. Lunner Kolstrup, D. I. Douphrate, S. Pinzke, and T. Grandin. 2013. Occupational health and safety aspects of animal handling in dairy production. *J. Agromedicine* 18:274–283. <https://doi.org/10.1080/1059924X.2013.796906>.
- Lobeck, K. M., M. I. Endres, E. M. Shane, S. M. Godden, and J. Fetrow. 2011. Animal welfare in cross-ventilated, compost-bedded pack, and naturally ventilated dairy barns in the upper Midwest. *J. Dairy Sci.* 94:5469–5479. <https://doi.org/10.3168/jds.2011-4363>.
- Lunner Kolstrup, C., M. Kallioniemi, P. Lundqvist, H. R. Kymäläinen, L. Stallones, and S. Brumby. 2013. International perspectives on psychosocial working conditions, mental health, and stress of dairy farm operators. *J. Agromedicine* 18:244–255. <https://doi.org/10.1080/1059924X.2013.796903>.
- McCurdy, S. A., and D. J. Carroll. 2000. Agricultural injury. *Am. J. Ind. Med.* 38:463–480. [https://doi.org/10.1002/1097-0274\(200010\)38:4<463::AID-AJIM13>3.0.CO;2-N](https://doi.org/10.1002/1097-0274(200010)38:4<463::AID-AJIM13>3.0.CO;2-N).
- Mendly-Zambo, Z. 2023. Field notes: Looking upstream at the farmer mental health crisis in Canada. Accessed May 24, 2024. <https://policyalternatives.ca/publications/reports/field-notes>.
- MHRC (Mental Health Research Canada). 2023. National Population Polling on Canadian Mental Health. Accessed May 24, 2024. <https://www.mhrc.ca/national-polling>.
- Polain, J. D., H. L. Berry, and J. O. Hoskin. 2011. Rapid change, climate adversity and the next ‘big dry’: Older farmers’ mental health. *Aust. J. Rural Health* 19:239–243. <https://doi.org/10.1111/j.1440-1584.2011.01219.x>.
- Posit Team. 2024. RStudio: Integrated development environment for R. Version 2024.9.1.394. Posit Software, PBC.
- Qi, X., X. Yao, X. Cong, S. Li, M. Han, Z. Tao, X. Yang, X. Qi, F. Shi, and S. Wang. 2024. Profile and risk factors in farmer injuries: A review based on Haddon matrix and 5 E’s risk reduction strategy. *Front. Public Health* 12:1322884. <https://doi.org/10.3389/fpubh.2024.1322884>.
- Qualtrics. 2005. Qualtrics software. May 2023. Qualtrics.
- Regier, D. A., W. E. Narrow, and D. S. Rae. 1990. The epidemiology of anxiety disorders: The epidemiologic catchment area (ECA) experience. *J. Psychiatr. Res.* 24:3–14. [https://doi.org/10.1016/0022-3956\(90\)90031-K](https://doi.org/10.1016/0022-3956(90)90031-K).
- Rocha, J., S. Oliveira, C. M. Viana, and A. I. Ribeiro. 2022. Climate change and its impacts on health, environment and economy. Pages 253–279 in *One Health*. Academic Press.
- Rook, K. S. 1987. Social support versus companionship: Effects on life stress, loneliness, and evaluations by others. *J. Pers. Soc. Psychol.* 52:1132–1147. <https://doi.org/10.1037/0022-3514.52.6.1132>.
- Rudolph, J. M., R. L. Berg, and A. Parsaik. 2020. Depression, anxiety and stress among young farmers and ranchers: A pilot study. *Community Ment. Health J.* 56:126–134. <https://doi.org/10.1007/s10597-019-00480-y>.

- Scott, A. J., T. L. Webb, M. Martyn-St James, G. Rowse, and S. Weich. 2021. Improving sleep quality leads to better mental health: A meta-analysis of randomised controlled trials. *Sleep Med. Rev.* 60:101556. <https://doi.org/10.1016/j.smrv.2021.101556>.
- Shortall, O., L. A. Sutherland, A. Ruston, and J. Kaler. 2018. True cowmen and commercial farmers: Exploring vets' and dairy farmers' contrasting views of 'good farming' in relation to biosecurity. *Sociol. Ruralis* 58:583–603. <https://doi.org/10.1111/soru.12205>.
- Smith, M. M., C. L. Simms, and J. D. Aber. 2017. Case study: Animal bedding cost and somatic cell count across New England dairy farms: Relationship with bedding material, housing type, herd size, and management system. *Prof. Anim. Sci.* 33:616–626. <https://doi.org/10.15232/pas.2016-01601>.
- Spengler, S. E., S. R. Browning, and D. B. Reed. 2004. Sleep deprivation and injuries in part-time Kentucky farmers: Impact of self-reported sleep habits and sleep problems on injury risk. *AAOHN J.* 52:373–382. <https://doi.org/10.1177/216507990405200905>.
- Spitzer, R. L., K. Kroenke, J. B. Williams, and B. Löwe. 2006. A brief measure for assessing generalized anxiety disorder: The GAD-7. *Arch. Intern. Med.* 166:1092–1097. <https://doi.org/10.1001/archinte.166.10.1092>.
- Statistics Canada. 2022. Canada's 2021 Census of Agriculture: A story about the transformation of the agriculture industry and adaptiveness of Canadian farmers. Accessed May 10, 2024. <https://www150.statcan.gc.ca/n1/daily-quotidien/220511/dq220511a-cansim-eng.htm>.
- Statistics Canada. 2023. Socioeconomic portrait of Canada's farm population, 2021. Accessed May 10, 2024. <https://www150.statcan.gc.ca/n1/pub/11-627-m/11-627-m2023044-eng.htm>.
- Sturgeon, R., and P. J. Morrisette. 2010. A qualitative analysis of suicide ideation among Manitoban farmers. *Can. J. Couns. Psychother.* 44:191–207.
- Tambadou, H., A. Le Heiget, B. Zwick, G. Dallago, B. Hagen, C. Winder, J. C. Plaizier, J. Kinley, K. Ominski, E. Pajor, H. Watson, and M. King. 2026. Not your average dairy farmer: Clustering dairy farmers of Western Canada and Ontario based on management practices and well-being. *J. Dairy Sci.* 109:7056–7069. <https://doi.org/10.3168/jds.2025-27820>.
- Thompson, R., B. N. Hagen, M. N. Lumley, C. B. Winder, B. Gohar, and A. Jones-Bitton. 2022. Mental health and substance use of farmers in Canada during COVID-19. *Int. J. Environ. Res. Public Health* 19:13566. <https://doi.org/10.3390/ijerph192013566>.
- Torske, M. O., B. Hilt, D. Glasscock, P. Lundqvist, and S. Krokstad. 2016. Anxiety and depression symptoms among farmers: The HUNT study, Norway. *J. Agromedicine* 21:24–33. <https://doi.org/10.1080/1059924X.2015.1106375>.
- Torvik, F. A., K. Rognmo, and K. Tambs. 2012. Alcohol use and mental distress as predictors of non-response in a general population health survey: The HUNT study. *Soc. Psychiatry Psychiatr. Epidemiol.* 47:805–816. <https://doi.org/10.1007/s00127-011-0387-3>.
- Tse, C., H. W. Barkema, T. J. DeVries, J. Rushen, E. Vasseur, and E. A. Pajor. 2018. Producer experience with transitioning to automatic milking: Cow training, challenges, and effect on quality of life. *J. Dairy Sci.* 101:9599–9607. <https://doi.org/10.3168/jds.2018-14662>.
- Udry, J. R. 1998. Why are males injured more than females? *Inj. Prev.* 4:94–95. <https://doi.org/10.1136/ip.4.2.94>.
- Vandekerckhove, M., and Y.-L. Wang. 2017. Emotion, emotion regulation and sleep: An intimate relationship. *AIMS Neurosci.* 5:1–17. 10.3934/Neuroscience.2018.1.1.
- Ventura, G., V. Lorenzi, F. Mazza, G. A. Clemente, C. Iacomino, L. Bertocchi, and F. Fusi. 2021. Best farming practices for the welfare of dairy cows, heifers and calves. *Animals (Basel)* 11:2645–2659. <https://doi.org/10.3390/ani11092645>.
- Virtanen, M., J. E. Ferrie, D. Gimeno, J. Vahtera, M. Elovainio, A. Singh-Manoux, M. G. Marmot, and M. Kivimäki. 2009. Long working hours and sleep disturbances: The Whitehall II prospective cohort study. *Sleep* 32:737–745. <https://doi.org/10.1093/sleep/32.6.737>.
- Walker, J. L., and L. J. Walker. 1988. Self-reported stress symptoms in farmers. *J. Clin. Psychol.* 44:10–16. [https://doi.org/10.1002/1097-4679\(198801\)44:1<10::AID-JCLP2270440103>3.0.CO;2-6](https://doi.org/10.1002/1097-4679(198801)44:1<10::AID-JCLP2270440103>3.0.CO;2-6).
- Wang, C., R. C. Colley, K. C. Roberts, J. P. Chaput, and W. Thompson. 2022. Sleep behaviours among Canadian adults: Findings from the 2020 Canadian Community Health Survey healthy living rapid response module. *Health Rep.* 33:3–14. <https://doi.org/10.25318/82-003-x202200300001-eng>.
- Watson, N. F., M. S. Badr, G. Belenky, D. L. Bliwise, O. M. Buxton, D. Buysse, D. F. Dinges, J. Gangwisch, M. A. Grandner, C. Kushida, R. K. Malhotra, J. L. Martin, S. R. Patel, S. F. Quan, E. Tasali, M. Twery, J. B. Croft, E. Maher, J. A. Barrett, S. M. Thomas, and J. L. Heald. 2015. Recommended amount of sleep for a healthy adult: A joint consensus statement of the American Academy of Sleep Medicine and Sleep Research Society. *J. Clin. Sleep Med.* 11:591–592. <https://doi.org/10.5664/jcsm.4758>.
- Wheeler, L., H. Reis, and J. Nezelek. 1983. Loneliness, social interaction, and sex roles. *J. Pers. Soc. Psychol.* 45:943–953. <https://doi.org/10.1037/0022-3514.45.4.943>.
- Wilkie, R. 2010. *Livestock/Deadstock: Working with Farm Animals from Birth to Slaughter*. Temple University Press.
- Wilton, B. 2019. Exploring the Connection between Mental Health and Farm Management. Farm Management Canada. Accessed Oct. 22, 2025. <https://fmc-gac.com/wp-content/uploads/2020/06/Environmental-Scan.pdf>.
- Xia, Q., D. Bertolino, M. S. Salim, L. V. Torian, and Y. Zheng. 2024. The case for retiring "period prevalence." *Int. J. Epidemiol.* 53:dyad186. <https://doi.org/10.1093/ije/dyad186>.

ORCIDS

- C. Winder, <https://orcid.org/0000-0002-7314-3657>
 J. C. Plaizier, <https://orcid.org/0000-0002-8551-9535>
 J. Kinley, <https://orcid.org/0000-0001-6853-1922>
 K. Ominski, <https://orcid.org/0000-0002-8047-8172>
 E. Pajor, <https://orcid.org/0000-0003-0747-3997>
 M. King <https://orcid.org/0000-0002-7562-2221>